

Mechanistic and kinetic insight using Kinetics-Informed Neural Networks (KINNs)

Developing an automated mechanistic elucidation tool for the Temporal
Analysis of Products (TAP) system

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Abstract

Heterogeneous catalysis is central to the chemical industry and will play a vital role in the transition from fossil-based feedstocks to renewable resources. Understanding catalytic performance requires mechanistic and kinetic insight, but conventional steady-state measurements often provide limited information about the elementary reaction steps. Transient techniques such as Temporal Analysis of Products (TAP) on the other hand, offer richer kinetic information. TAP is defined by the simplification of gas transport under low-pressure Knudsen diffusion conditions. This thesis explores the use of machine learning (ML) for the analysis of transient kinetic data, with a focus on extending TAP analysis using Kinetics-Informed Neural Networks (KINNs). KINNs incorporate governing kinetic equations into neural network training and can be used to extract kinetic parameters from transient data. Building on previous work applying KINNs to TAP, this project investigates whether the framework can be extended toward learning both kinetic parameters and reaction mechanisms from a larger reaction space. The proposed approach aims to move beyond testing fixed mechanisms one by one by introducing a framework for automated mechanistic exploration. Our investigations reveal remaining challenges when it comes to the application of KINNs to mechanistic and kinetic elucidation.

Preface

I would first like to thank my supervisors for their guidance, support and feedback throughout this project. Their encouragement have been important in shaping both the direction of the work and my understanding of the topic.

I would also like to thank the Department of Chemistry for providing a good academic environment and the resources needed to carry out this thesis. As well as for funding my attendance at a workshop relevant to the thesis topic where I had the opportunity to present my first poster.

Finally, I would like to thank my family and friends for their patience, encouragement, and support throughout my studies.

Hanne Jeeyung Eriksen
Oslo, Spring 2026

Deceleration

In this scientific work, generative artificial intelligence (AI) has been used. All data and personal information have been processed in accordance with the University of Oslo's regulations, and I, as the author of the document, take full responsibility for its content, claims, and references. An overview of the use of generative AI is provided below.

The service ChatGPT-4, developed by OpenAI, has been used while 1) working on the thesis as a discussion partner to deepen understanding of topics as well as scanning-like tasks such as error searching of uploaded original code in certain cases, and 2) while writing this document to improve formulation of sentences and grammar for readability. Beyond the sparring in discussion, the model has not generated something that wasn't based on original work/formulations.

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Acronyms

BC Boundary condition

ER Eley-Rideal

FFNN Feedforward neural network

IC Initial condition

KINN Kinetics-informed neural network

LH Langmuir-Hinshelwood

MKM microkinetic model

ML Machine Learning

MSE mean squared error

ODE Ordinary differential equation

PDE Partial differential equation

PINN Physics-informed neural network

STE straight-through estimator

TAP Temporal Analysis of Products

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Part I

Introduction

Chapter 1

Introduction of concepts

Catalysis is the acceleration of a chemical reaction by providing an alternative reaction pathway with a lower activation energy. Catalytic processes are central to modern chemical industry, where they are used in processes such as ammonia synthesis, catalytic cracking, hydrogenation, and emission control.[22, 33, 43] A particularly important class is heterogeneous catalysis. Here, the catalyst and reactants are in different phases, most commonly a solid catalyst in contact with gaseous or liquid reactants. The reaction therefore takes place at the interface between the phases, usually on the catalyst surface. Heterogeneous catalysts are often preferred because they can be separated from the reaction mixture, reused, and implemented in continuous reactor systems. For this reason, heterogeneous catalysis remains a central technology in the chemical industry and is expected to play an important role in the transition from fossil-based feedstocks to renewable resources.[22]

As catalysts facilitate reactions by lowering activation energy barriers, catalysis research is fundamentally tied to chemical kinetics, the study of reaction rates. The precise determination of reaction mechanisms and kinetic coefficients is essential for understanding and improving catalytic performance.[3, 31] Whereas the simpler and more commonly used steady-state techniques can offer a global snapshot of catalyst performance, and are therefore useful for catalyst screening, transient methods are needed to reveal deeper information about the underlying reaction mechanism.[2, 11] These methods are based on introducing a perturbation to the system and monitoring the time-dependent response caused by a rapid change in a reaction variable, such as concentration or temperature.[2] For instance, in the Temporal Analysis of Products (TAP) pulse method [15, 47], a short and sharp gas pulse is introduced into a microreactor, followed by recording of the outlet flux as a function of time. This makes TAP a useful technique for studying catalytic kinetics under transient conditions. TAP experiments have been used to elucidate the kinetics of reactions such as propane dehydrogenation,[57] methanol oxidation,[41] and many others.[36] Although traditional kinetic data analysis provides essential tools for estimating parameters and validating reaction models, the interpretation of increasingly complex catalytic systems can be non-trivial, labor-intensive, and computationally costly. In this context, machine learning can serve as a complementary or alternative approach.[52]

Machine Learning (ML) methodologies have become firmly established in the natural sciences over the past decade.[6] In chemistry, these approaches have enabled a wide range of applications, spanning property prediction, reaction modeling, and inverse molecular design.[4] A significant portion of this research has focused on strategies for using large datasets to model and screen thousands of molecules, particularly in

the context of drug discovery.[50] More recently, increasing attention has been directed toward ML approaches that incorporate mechanistic chemical knowledge directly into the learning process. This is especially valuable in low-data regimes. For example, Physics-informed neural networks (PINNs) integrate physical and chemical constraints, often expressed as differential equations, into neural-network architectures.[9] By embedding mechanistic information into the optimization process, these models have emerged as promising tools for describing complex chemical systems and reaction dynamics in a physically consistent manner.[17]

In the TAP community, ML has not yet received the same level of attention as in many other fields. This is highlighted by Wang et al., who discuss the need to make TAP data more available to data scientists in order to improve the efficiency, quality, and amount of information that can be extracted from traditional TAP data analysis.[52] In addition, low-data methods are especially relevant for catalytic experiments, since collecting high-quality reaction data can be expensive and labor-intensive. One area where ML has already been successfully applied to TAP is through the use of Kinetics-informed neural networks (KINNs).[17] KINNs are related to PINNs, but are specifically formulated for kinetic systems. Nai et al. applied KINNs to TAP data for the extraction of rate constants and interpolation between pulses.[34] Their results show that KINNs can connect transient TAP data with kinetic parameter estimation, providing a basis for exploring whether the framework can be extended toward richer mechanistic inference. A remaining limitation is that the mechanism must still be assumed before training. In practice, this means that different candidate mechanisms must be tested one by one, which can become a large and computationally demanding search space. This challenge becomes especially important for reaction systems with many possible elementary steps, such as combustion reaction networks, where stiff kinetic equations can also lead to high computational cost.[5]

Chapter 2

Aim and outlook of thesis

The aim of this project is to develop a machine learning model for TAP analysis that expands upon existing methods in a field where machine learning is still not fully leveraged.

After introducing the theoretical background of the methods used, the proposed model is presented together with the data. The model is then evaluated using a reference case before different aspects of the framework are tested to study their effect on performance and to gain a deeper understanding of the model. Finally, the thesis concludes with a critical evaluation of the employed methods and perspectives for future work.

A further aim of the thesis is to make the work accessible to both chemists and computational scientists by building up the necessary concepts from both disciplines from the bottom.

Part II

Theoretical background

Chapter 3

Heterogeneous catalysis

Catalysts are used to make chemical reactions more efficient and controllable as many reactions occur too slowly or requires harsh conditions. A catalyst can improve this by increasing the reaction rate and/or guide the reaction toward a specific product. The goal of a catalyst is therefore to increase the activity and/or selectivity of a reaction. Activity is related to how fast the reaction proceeds, and an important metric is the turnover frequency (TOF) which describes how many molecules are converted per active site per unit time. Selectivity, on the other hand, describes how much of the converted reactant forms the desired product instead of unwanted by-products.[35] These quantities are important in catalysis research because they make it possible to compare catalyst performance, but what causes these measured quantities is the underlying mechanism. Understanding this mechanism is needed to create better catalysts, and is therefore a major focus in catalysis research.[20]

As a basic definition, consider a reaction where species A reacts with species B to form product P with their respective stoichiometric coefficients v_i



An example of such a reaction is CO oxidation $\text{CO} + \frac{1}{2}\text{O}_2 \rightarrow \text{CO}_2$. This type of reaction can be sped up with the help of a catalyst. The catalyst provides an alternative reaction pathway with a lower activation energy, making it easier for the reactants to reach the transition state and form products. In this way, the catalyst affects the kinetics of the reaction, while the overall thermodynamic equilibrium remains unchanged. In heterogeneous catalysis, the catalyst is in a different phase than the reactants, usually as a solid with gas or liquid reactants. The reaction therefore takes place at the surface of the catalyst.[35] The catalyst can be porous or non-porous, and porous catalysts are often useful because they give a larger surface area where the reaction can occur.[48] Many catalysts are also supported, meaning that the active material is dispersed on another material. The support can stabilize the active phase, improve dispersion, and in some cases also affect the reaction mechanism. Designing catalysts is therefore closely connected to understanding which surface sites are active and which mechanism the reaction follows. [32]

The above reaction is an example of an overall reaction, this means that it only shows the total conversion from reactants to products, but not the actual sequence of steps. In reality, the reaction normally happens through several elementary steps coupled together. An elementary reaction is a singular molecular event such as for example an adsorption of CO onto the surface. Each elementary step has its own transition state and activation barrier, and the full mechanism is the sequence of these

steps. The elementary reactions usually involve only one or two individual species. The chance that they occur depends on the amount of the reacting species, for example their concentration or surface coverage, and on whether they have enough energy to overcome the activation barrier. The catalytic cycle in heterogeneous catalysis usually starts with adsorption of reactants on the catalyst surface. Adsorption can be divided into physisorption and chemisorption. In physisorption, the molecule interacts weakly with the surface, mainly through van der Waals forces. In chemisorption, the molecule forms a stronger chemical bond with the surface. Adsorption is important because it brings the reactants to the surface and can activate them for further reaction.[35] In general, heterogeneous catalysis works by the surface activating the molecules. This means that the molecules adsorb onto the surface and react in ways they would not easily do by themselves, especially for molecules with strong bonds such as N_2 . A useful way to construct possible elementary steps is to consider how the reactants adsorb and are activated on the surface, how they may break into smaller surface species, and how these species can combine again to form the product. Either one reactant can adsorb and react with another reactant directly from the gas phase, which is called the Eley-Rideal (ER) mechanism, or both reactants can adsorb onto the surface and react there, which is referred to as the Langmuir-Hinshelwood (LH) mechanism. Another possibility is the Mars-van Krevelen mechanism, where the catalyst itself participates in the reaction, for example by providing lattice oxygen that is later regenerated.[24]

Determining the exact mechanism, or the exact sequence of elementary steps, is no easy task. It becomes complicated because several pathways may be possible at the same time, and different pathways can sometimes give similar overall behavior. The mechanism can also depend on temperature, pressure, surface coverage, and the state of the catalyst. In addition, the support can also be involved in the mechanism, and the reaction may be structure dependent. For example, the type of site can matter, since undercoordinated sites are often more reactive than terrace sites. Yet another complicating factor is adsorbate-adsorbate interactions. The adsorbed species on the catalyst surface can interact with each other, which can lead to patterns or islands on the surface. These interactions change the local environment of the active sites and can affect adsorption energies, activation barriers, and competition for sites. As a result, they can influence both the rate and the mechanism of the reaction. [35] For example in CO oxidation, CO and O_2 can compete for the same limited surface sites, which can cause the adsorbed species to separate into CO rich and oxygen rich regions. Since the reaction then mainly occurs where these regions meet, the surface organization itself influenced the observed reaction rate and product formation. [29] Furthermore, there is the lifetime of the catalyst to consider. The state of the catalyst can change over the course of a reaction, for example from oxidized to reduced or from fresh to poisoned. Poisoning can be chemical, thermal, or mechanical, and can include processes such as sulfur poisoning, coking, sintering, or pore blocking. This can change how the catalyst behaves and may also change the reaction mechanism. The stability metric turnover number (TON) is thus another common metric in catalysis research.[35] Overall, these factors show why mechanism determination in heterogeneous catalysis is challenging. The observed reaction behavior may depend not only on the elementary steps themselves, but also on the catalyst structure, adsorbate interactions, support effects, and changes in catalyst state. This complexity motivates the need for methods that can extract mechanistic information from experimental data and help distinguish between possible reaction pathways.[49]

Chapter 4

Chemical kinetics

Kinetic analysis is central to mechanism determination, since it connects experimental reaction data with possible mechanistic explanations. Kinetic characterization involves collecting reaction data and interpreting it using mathematical models and data analysis. In this chapter the basics of kinetic modeling, data analysis, and chemical reactors will be introduced, with an emphasis on the TAP reactor which is central to this thesis.

4.1 Kinetic modeling

4.1.1 Kinetics of elementary reactions

To express an overall reaction rate, the simplest approach is to set up a rate based on a power law where the exponents equal to the stoichiometric coefficients in the reaction equation. For example for the overall reaction in eq. 3.1

$$r = k[A]^{v_A}[B]^{v_B} \quad (4.1)$$

That being said, experiments have shown that this approach is often too simple as it rare for the reactant stoichiometric coefficients to be the same as the exponents in these power laws. This stems from the fact that such power laws do not properly describe the kinetics of the reaction as this requires describing the elementary reaction steps that make up the overall reaction. [35]

For a given elementary reaction j , the reaction rate r_j ($\text{molm}^{-2}\text{s}^{-1}$) can be expressed as a power law expression involving a rate constant k_j given by

$$r_j = k_j \prod \theta^{v_k} C_g^{v_i} \quad (4.2)$$

where now the gas is expressed in concentration, here denoted as C_g ($\text{molm}^{-3}\text{s}^{-1}$), and the adspecies are expressed in fractional coverage θ_i (unitless) as given in the following equation calculated from surface concentration C_s ($\text{molm}^{-2}\text{s}^{-1}$)

$$\theta_i = \frac{C_{s,i}}{C_{s,*,tot}} \quad (4.3)$$

where $C_{s,i}$ is surface concentration for a given species i and $C_{s,*,tot}$ is the total concentration of active sites. Furthermore, the total number of surface sites is conserved

$$\theta_i + \dots + \theta_* = 1 \quad (4.4)$$

For instance, the rate law of an adsorption reaction $A + * \rightarrow A^*$ is expressed by

$$\begin{aligned} r_a &= k_a \cdot C_{g,A} \cdot \theta_* \\ &= k_a \cdot C_{g,A} \cdot (1 - \theta_A) \end{aligned} \quad (4.5)$$

The intuition/interpretation behind these rate laws lies in that coverages can be interpreted as probabilities. For example, for the reaction just mentioned, the rate law is then saying that the rate at which adsorption occurs on a given site on the surface is proportional to the product of the rate constant and the probability that the site is unoccupied θ_* . Following this, when more than two sites are involved, for example in associative desorption $2A^* \rightarrow A_2 + 2^*$ with the rate law

$$r_d = k_d \cdot \theta_A^2 \quad (4.6)$$

Here the squared coverage θ_A^2 can then be thought of as describing the probability of finding two sites both occupied with A next to each other. [35]

The rate is furthermore proportional to the rate constant, which is exponentially depended on temperature T following the empirical Arrhenius law or alternatively the theoretical Eyring equation resulting from transition state theory given by

$$k = \frac{k_B T}{h} \exp\left(-\frac{\Delta G^\ddagger}{RT}\right) \quad (4.7)$$

where k_B is the Boltzmann constant, h is Planck constant, R is the universal gas constant and $\Delta G^\ddagger$ is the Gibbs free energy of activation. [27]

4.1.2 Microkinetic modeling

The rate laws above describe individual elementary reaction steps and these steps are connected through shared gas and surface species in a given reaction pathway. This means that the rate of one species may depend on several elementary reactions. A microkinetic model (MKM) combines these elementary steps into a set of coupled Ordinary differential equations (ODEs) describing how the concentrations and coverages in the system evolve with time. [40]

Developing a MKM firstly begins by proposing a plausible reaction mechanism. Rate expressions are then written for each elementary reaction as described in eq. 4.2. These rate expressions are then used to construct a system of ODEs, where each equation describes the overall transformation rate of a species in the mechanism. Here the overall transformation rate for a given species is obtained from a balance over all elementary steps involving that species. For a gas species A this can be written as

$$R_A = \frac{\partial C_A}{\partial t} = C_{s,*,tot} \sum_i \nu_{A,i} r_i \quad (4.8)$$

where C_A is the gas concentration, $C_{s,*,tot}$ is a scaling factor equal to the total concentration of active surface sites, $\nu_{A,i}$ is the stoichiometric coefficient of species A in elementary step i , and r_i is the rate of elementary step i . For a surface species A^* the equivalent expression is

$$R_{A^*} = \frac{\partial \theta_A}{\partial t} = \sum_i \nu_{A^*,i} r_i \quad (4.9)$$

[35] The same system can be written more compactly by introducing the state vector

$$\mathbf{x} = \begin{bmatrix} \mathbf{c}_g \\ \boldsymbol{\theta} \end{bmatrix} \quad (4.10)$$

where $\mathbf{c}_g$ contains the gas concentrations and $\boldsymbol{\theta}$ contains the surface coverages. The MKM can then be written as

$$\dot{\mathbf{x}} = \mathbf{M}\mathbf{r}(\mathbf{c}_g, \boldsymbol{\theta}, T) = \mathbf{M}(\mathbf{k}(T) \odot \boldsymbol{\psi}(\mathbf{c}_g, \boldsymbol{\theta})) \quad (4.11)$$

Here $\mathbf{M}$ is the stoichiometric matrix, $\mathbf{r}$ is the vector of elementary reaction rates, $\mathbf{k}(T)$ is the vector of Arrhenius temperature dependent rate constants, and $\boldsymbol{\psi}(\mathbf{c}_g, \boldsymbol{\theta})$ maps the concentration and coverages to the corresponding power law. The symbol $\odot$ denotes element-wise multiplication. [17] In general, the resulting system of ODEs cannot be solved analytically and must instead be solved numerically.

This formulation is based on the mean-field approximation where the catalyst surface is described by average coverages rather than the explicit positions of individual adsorbates. This can introduce errors when such spatial effects become important. More extensive methods such as kinetic Monte Carlo simulations can include such effects, but at a higher computational cost compared to the cheap MKM.[37]

4.2 Chemical reactors and transient techniques

A chemical reactor is the unit where the feed is converted into the desired products. The design and operation of the reactor are therefore central to chemical processes, since they determine how transport and reaction occur together in the system. [30]

For the modeling of any chemical reactor, the basic balance equation can be written as

$$\text{accumulation} = \text{transport in} - \text{transport out} + \text{reaction term} \quad (4.12)$$

Based on conservation of mass, this states that the rate of accumulation inside a reactor volume element is equal to the rate of flow into the volume element minus the rate of flow out, plus a reaction source term. [28] The reaction term contains the kinetic and mechanistic description of the system through the MKM, as described in Eq. 4.11. Reactor data therefore contains a combination of transport and reaction effects which must be separated in order to extract kinetic information.

In steady-state experiments the system is studied after the reactor has reached conditions where the measured quantities no longer change with time. Steady-state techniques, such as plug flow reactors and continuously stirred tank reactors, are widely used because they are relatively simple and useful for evaluating catalyst performance. However the low time resolution limits the amount of mechanistic information that can be extracted. These methods can provide global information about conversion, yield, and selectivity, but more detailed methods are needed to study the elementary steps of a microkinetic mechanism. [2, 11] Transient techniques instead study the response of the system after a rapid change in a reaction variable. This can for example be a change in concentration, temperature, pressure, or flow rate. The time dependent response contains information about the underlying kinetics because the system is followed while it relaxes from one state to another. [2]

For transient experiments the reactor model generally has to describe both time and space. In contrast to the ODEs used for the MKM alone, reactor models are often written as Partial differential equations (PDEs), since the concentration may vary with

both position and time. The reaction term in the reactor model is still given by the MKM, but it is now coupled to transport in the reactor. To extract kinetic parameters from transient reactor data, a candidate mechanism is first assumed. This determines the reaction term through the rate expressions and the stoichiometric matrix. The resulting reactor model is then solved numerically for a given set of kinetic parameters. The simulated concentration or flux data are compared with the experimental data, and the kinetic parameters are adjusted until the difference is minimized. This can be written as

$$\min_{\mathbf{k}} \sum_{n=1}^N \left(C_n - \hat{C}_n(\mathbf{k}) \right)^2 \quad (4.13)$$

where C_n is the experimental data point, $\hat{C}_n(\mathbf{k})$ is the simulated value obtained using the kinetic parameters $\mathbf{k}$, and N is the number of datapoints. The optimization can be carried out using different algorithms including direct search methods and gradient-based methods. In practice several candidate mechanisms may have to be tested. This procedure is summarized in Fig. 4.1. More details on this approach can be found in [51].

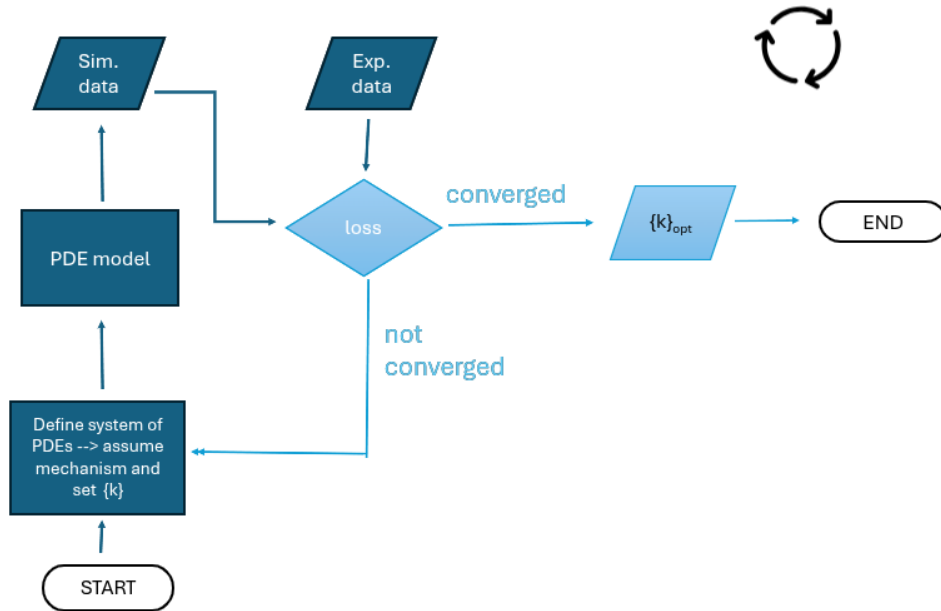


Figure 4.1: Flowchart for extraction of kinetic and mechanistic properties from transient experimental data

4.3 TAP

An important transient technique is the TAP reactor [15, 47]. The method can be used to measure quantities that are important for catalyst characterization such as the number of active sites, sticking coefficients, and mechanistic features obtained through numerical fitting. [12]

4.3.1 Basic concepts and experimental set up

The TAP reactor consists of a high-speed pulse valve which delivers gas into a packed-bed microreactor. A mass spectrometer is placed at the reactor outlet for detection

as shown in figure 4.2. The packed-bed reactor can have different configurations with zones of either active catalyst material or inert particles such as quartz. The simplest configuration is a single active zone. A more common configuration is a three-zone reactor where an active zone is surrounded by inert zones.[14] A useful requirement in kinetic catalyst characterization is uniformity of the reactant concentration profile within the active catalyst zone. This can be achieved either by forced mixing or by making the active zone thin.[45] A commonly used configuration based on this is the thin-zone TAP reactor. This can be seen as a limiting case of the three-zone model where the active zone is sufficiently thin to reduce gradients across the catalyst zone.[47]

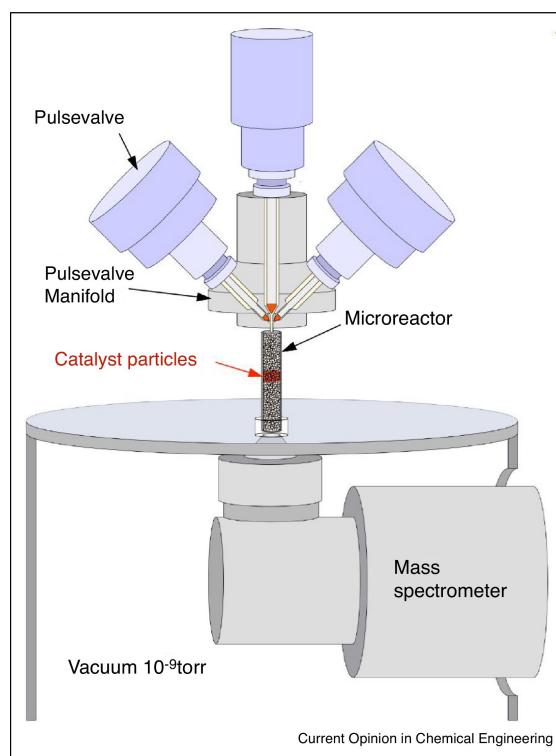


Figure 4.2: Schematic overview of main components of the TAP instrumentation. Taken from [13].

A defining characteristic of TAP experiments is the use of ultralow gas pulse intensities of around 10 nmol per pulse. In TAP experiments transport is not eliminated as in some steady-state kinetic measurements. Instead gas transport occurs in the well defined Knudsen diffusion regime, due to the ultralow intensity, which makes it possible to separate transport effects from the reaction information in the measured response.[14] The small pulse furthermore only weakly perturbs the catalyst state meaning that a pulse can therefore give a kinetic snapshot of the system. These experiments are often referred to as state-defining experiments. Another mode is state-altering experiments where the catalyst state changes over time. This can for example be done through multiple pulses where each individual pulse is small but the accumulated effect changes the catalyst surface. Such experiments can be used to follow changes in catalyst behavior, for example from a fresh to a poisoned state or from an oxidized to a reduced state.[12] At the reactor exit, the outlet flux is measured by the mass spectrometer. If only diffusion occurs, the outlet response follows a diffusion curve. If reaction also occurs, the outlet response changes. This change contains information about the reaction. The concepts of insignificant perturbation and diffusional flow are therefore central to what distinguishes

the TAP reactor from many other kinetic tools.[14]

4.3.2 Modeling and data analysis

The most straightforward quantitative TAP analysis is the numerical fitting approach used to discriminate and validate different MKMs. In this approach a reactor model is solved and the predicted outlet response is compared with the measured outlet flux. For the active zone of the TAP reactor, the balance equation introduced in Eq. 4.12 can be written for a gas species A as

$$\epsilon_b \frac{\partial C_{g,A}}{\partial t} = D_{eA} \frac{\partial^2 C_{g,A}}{\partial x^2} - (1 - \epsilon_b) a_v R_A \quad (4.14)$$

where ϵ_b is the bed void fraction, a_v is the specific surface area and D_{eA} is the effective Knudsen diffusivity. The reaction source or sink term is given by R_A as defined in Eq. 4.8. In the inert zones there is no reaction $R_A = 0$ which corresponds to Fick's second law of diffusion under characteristic TAP Knudsen transport.[12] For a surface species A^* there is no transport and the balance is written as

$$\frac{\partial \theta_A}{\partial t} = R_{A^*} \quad (4.15)$$

where R_{A^*} is the rate of change of the surface coverage as defined in Eq. 4.9. For the Initial condition (IC) belonging to the system of PDEs, the reactor is taken to be fully evacuated at the beginning of the experiment. The initial gas concentration is therefore

$$C_{g,A}(x, 0) = 0 \quad (4.16)$$

For surface species, the initial condition depends on the catalyst state. A surface species may also be absent at the start of the experiment or it can be present due to pretreatment or previous pulses. This can be written as

$$\theta_A(x, 0) = \theta_{A,init} \quad (4.17)$$

Furthermore, for the Boundary condition (BC) of the gas species, the reactor inlet is specified at $x = 0$ for

$$\frac{\partial C_{g,A}(0, t)}{\partial x} = N_p \frac{t}{\tau^2} \exp\left(-\frac{t}{\tau}\right) \quad (4.18)$$

where N_p is the amount of gas in the pulse and τ is the tortuosity of the reactor bed. On the other hand at the reactor outlet $x = L$, the boundary condition is

$$C_{g,A}(L, t) = 0 \quad (4.19)$$

where L is the reactor length. For a three-zone reactor, internal boundary conditions are also needed between the zones. These enforce continuity of concentration and flux across the zone boundaries. Lastly, the flux exiting the reactor is defined as

$$F_{A,exit} = -D_{eA} \frac{\partial C_{g,A}(L, t)}{\partial x} \quad (4.20)$$

An overview of the full model can be seen in Fig. 4.3. [14]

This approach is model-based, meaning that it includes prior assumptions about the reaction mechanism through the rate term. Because of this, model-free approaches have also been developed to extract information from the data without assuming a full

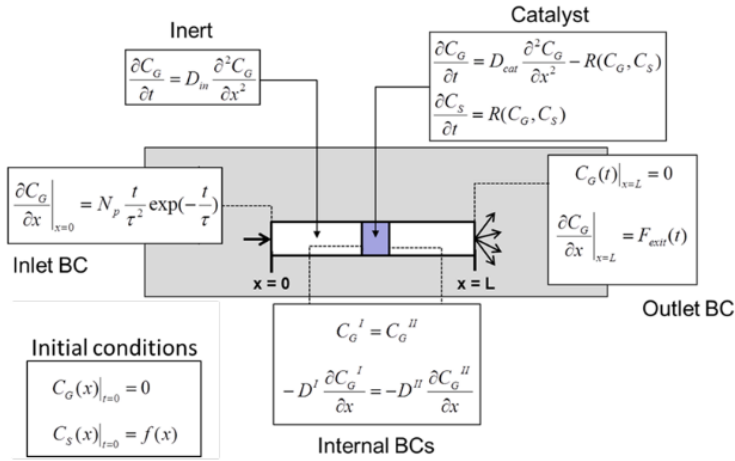


Figure 4.3: Full PDE-based model used for numerical regression.

mechanism. This includes Shekhtman reactivities, which are integral quantities obtained from the outlet flux curves.[46] Beyond integral quantities, time-dependent analysis can preserve more of the transient information. For the thin-zone configuration, the catalyst zone can be treated as a thin reactive interface between two inert zones. This enables a physical decoupling of transport and kinetic processes. Transport information can be decoupled while keeping the time dependence by subtracting the response of the same molecule through an inert bed or by scaling an inert gas pulse according to Graham's law.[12] For the thin-zone configuration, the uniformity of the catalyst zone also allows the use of the Y-procedure. This procedure translates exit flow rates into net production rates and gas concentrations inside the uniform catalyst zone.[53] The procedure is based on Fourier-domain analysis of the reaction-diffusion equations used to describe TAP experiments.[8]

Chapter 5

Machine learning

Moving on to the second fundamental aspect of this thesis from the world of chemistry to ML, this section starts by introducing the basics and mathematical formulation of a very fundamental ML approach before finally arriving at the method of focus in this work.

5.1 Feedforward neural networks

A popular workhorse in ML is Feedforward neural network (FFNN)s, which are artificial neural networks inspired by biological neural networks. They are built from simple connected units called neurons which form a network that can learn relationships between inputs and outputs. Furthermore, in a FFNN information only flows in one direction from the input layer to the output layer. In other words, the connections do not loop back to earlier nodes. This separates FFNNs from for example recurrent neural networks. The neurons are structured in layers, and in a so called "fully connected neural network", each neuron is connected to neurons in adjacent layers but not to neurons within the same layer. A given connection between two neurons is associated with a weight which controls how much one neuron contributes to a neuron in the following layer. [16]

Mathematically a FFNN, taking an input $\mathbf{x} \in \mathbb{R}^{D_0}$ and H hidden layers, can be given by $\mathbf{f}(\mathbf{x}, \boldsymbol{\omega}_{ff}) : \mathbb{R}^{D_0} \rightarrow \mathbb{R}^{D_{H+1}}$ given by eq. 5.1, where the j th layer has width or dimensionality of D_j for $j \in \{0, \dots, H+1\}$.

$$\mathbf{f}(\mathbf{x}, \boldsymbol{\omega}_{ff}) = \boldsymbol{\sigma}_{H+1} \circ \mathbf{A}_{H+1} \circ \boldsymbol{\sigma}_H \circ \mathbf{A}_H \circ \dots \circ \boldsymbol{\sigma}_1 \circ \mathbf{A}_1(\mathbf{x}) \quad (5.1)$$

Here, for any $1 \leq j \leq H+1$, the affine linear transformation $\mathbf{A}_j(\mathbf{y}) : \mathbb{R}^{D_{j-1}} \rightarrow \mathbb{R}^{D_j}$ is

$$\mathbf{A}_j(\mathbf{y}) = \mathbf{W}_j \mathbf{y} + \mathbf{b}_j \quad (5.2)$$

where $\mathbf{W}_j \in \mathbb{R}^{D_j \times D_{j-1}}$ is the weight matrix, $\mathbf{b}_j \in \mathbb{R}^{D_j}$ is the bias vector, and $\boldsymbol{\omega}_{ff} = \{\mathbf{W}_j, \mathbf{b}_j \mid j \in \{1, \dots, H+1\}\}$. With this a FFNN can approximate functions mapping input to output, which is the basis for using neural networks as general function approximators. [10, 19] An intermediate result is often denoted as $\mathbf{a}^{(j)}$, where the linear combination before the activation function is given by $\mathbf{z}^{(j)} = \mathbf{A}_j(\mathbf{a}^{(j-1)})$. This gives

$$\mathbf{a}^{(j)} = \boldsymbol{\sigma}_j(\mathbf{z}^{(j)}) = \boldsymbol{\sigma}_j(\mathbf{A}_j(\mathbf{a}^{(j-1)})) \quad (5.3)$$

where $\mathbf{a}^{(0)} = \mathbf{x}$. Furthermore $\boldsymbol{\sigma}_j(\mathbf{z}^{(j)}) : \mathbb{R}^{D_j} \rightarrow \mathbb{R}^{D_j}$ is an activation function which introduces non-linearity into the network, allowing the model to represent more complex

relationships in the data. The functions can vary from layer to layer and there are many activation functions available. One very popular example is the rectified linear unit ReLU which zeros negative values

$$\text{ReLU}(\mathbf{z}^{(j)}) = \max(\mathbf{0}, \mathbf{z}^{(j)}) \quad (5.4)$$

[18]

During training the network aims to minimize a predefined loss function

$$\min_{\omega_{ff}} \mathcal{L}(\omega_{ff}) \quad (5.5)$$

The loss function is often represented as the mean squared error (MSE) for regression problems or cross-entropy for classification tasks. For example in the case of MSE, the loss function can be written for N datapoints as

$$\mathcal{L}_{\text{MSE}}(\omega_{ff}) = \frac{1}{N} \sum_{n=1}^N \left(f(\mathbf{x}_n, \omega_{ff}) - \tilde{f}_n \right)^2 \quad (5.6)$$

where $f(\mathbf{x}_n, \omega_{ff})$ is the network prediction for datapoint n , $\tilde{f}_n$ is the corresponding target value, and N is the number of datapoints. [18] This is achieved through optimizers which are algorithms that update the trainable parameters of the model. One of the most common algorithms used to optimize neural networks, as well as the foundational algorithm for many other optimizers is gradient descent [42], given by

$$W_{ik, \text{new}} = W_{ik, \text{old}} - \alpha \frac{\partial \mathcal{L}}{\partial W_{ik, \text{old}}} \quad (5.7)$$

Like the name suggests, gradients are used to update the parameters as the gradients indicate how the weights should be changed in order to reduce the loss function. The extent of this update is controlled by the learning rate α . In regards to the calculation of the gradients, the fundamental backpropagation algorithm [44] works by moving backward from the model output. Backpropagation applies the chain rule to calculate how changes in each neural network parameter affect the overall prediction error. This is an efficient way to calculate gradients because intermediate derivatives can be reused as the gradients are propagated backward through the network. This is then a type of automatic differentiation (AD). To illustrate this for the simplest FFNN with one node in the input, hidden $a_1^{(1)}$ and output $a_1^{(2)}$ layers, the weights are $W_{11}^{(0)}$ and $W_{11}^{(1)}$. Here the superscript refers to the layer while the subscripts refer to the connected neurons. With backpropagation, the two weight derivatives can thus be expressed as

$$\frac{\partial \mathcal{L}}{\partial W_{11}^{(1)}} = \left(\frac{\partial \mathcal{L}}{\partial a_1^{(2)}} \right) \left(\frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \right) \left(\frac{\partial z_1^{(2)}}{\partial W_{11}^{(1)}} \right) \quad (5.8)$$

$$\frac{\partial \mathcal{L}}{\partial W_{11}^{(0)}} = \left(\frac{\partial \mathcal{L}}{\partial a_1^{(2)}} \right) \left(\frac{\partial a_1^{(2)}}{\partial z_1^{(2)}} \right) \left(\frac{\partial z_1^{(2)}}{\partial a_1^{(1)}} \right) \left(\frac{\partial a_1^{(1)}}{\partial z_1^{(1)}} \right) \left(\frac{\partial z_1^{(1)}}{\partial W_{11}^{(0)}} \right) \quad (5.9)$$

For a more general FFNN, a given weight can affect the loss through all later layers. For a given weight $W_{ik}^{(j)}$ the derivative can be written as the following compact form of the chain rule

$$\frac{\partial \mathcal{L}}{\partial W_{ik}^{(j)}} = \left(\frac{\partial \mathcal{L}}{\partial z_i^{(j+1)}} \right) \left(\frac{\partial z_i^{(j+1)}}{\partial W_{ik}^{(j)}} \right) = \frac{\partial \mathcal{L}}{\partial z_i^{(j+1)}} a_k^{(j)} \quad (5.10)$$

where the last derivative can be written as $a_k^{(j)}$ since

$$z_i^{(j+1)} = \sum_{k=1}^{D_j} W_{ik}^{(j)} a_k^{(j)} + b_i^{(j)} \quad (5.11)$$

After the calculation of these gradients, the weights are then updated as shown in the gradient descent equation. Within gradient descent the dataset can either be transversed in full or in terms of smaller batches of the dataset. Moreover, a popular upgraded optimizer based on gradient descent is Adam [25] which utilizes individual adaptive learning rates for different parameters.

For the evaluation of regression models, it is normal to use the mean absolute error (MAE) in addition to the MSE, as this is more directly interpretable in the same units as the output. However the MSE is often preferred during backpropagation because it is differentiable and penalizes larger errors more strongly. The MAE is given by

$$\text{MAE} = \frac{1}{N} \sum_{n=1}^N |f(\mathbf{x}_n, \boldsymbol{\omega}_{ff}) - \tilde{f}_n| \quad (5.12)$$

Another useful metric is R^2 , which measures how much of the variance in the target data is explained by the model with values ranging from 0 to 1. [18]

5.2 Kinetics-informed neural networks

Unlike the FFNNs discussed in the previous section, which are purely data-driven models, PINNs [38] take into account the underlying PDE or ODE describing the physical problem for example newton's law of cooling. In this way, the neural network is not only fitted to data, but is also constrained to satisfy the physical law of the system. PINN are useful when the system is governed by differential equations but the available data is limited, noisy, or incomplete. The FFNN type acts as an approximator for the unknown solution, while the governing equation is enforced through the loss function as a separate term alongside the standard data loss term. Training then involves a balance between fitting the observed data and satisfying the physical model.

Modified PINNs, called KINNs [17], were later introduced by Gusamo et al. for solving kinetic ODEs. In other words, the model is constrained by the kinetic equations from a defined MKM. A KINN can be summarized as two parts:

1. A FFNN, which acts as an approximator for the solution of the kinetic ODEs.
2. A loss function, which contains both a data loss term and a physics loss term.

There are furthermore two flavors of a KINN. In the forward problem, the kinetic parameters are assumed to be known and the FFNN is trained to solve the kinetic ODEs. In the inverse problem on the other hands, the kinetic parameters are unknown and are learned from data. The inverse KINN can therefore be seen as an extension of the forward KINN, where the goal is not only to fit the concentration profiles, but also to estimate the kinetic parameters.

Having defined a FFNN in Section 5.1, the KINN can be written as a mapping from time to concentration $\mathbf{c}(t) \approx \mathbf{f}(t, \boldsymbol{\omega}_{ff})$ where t is time, $\boldsymbol{\omega}_{ff}$ denotes the trainable FFNN weights, and $\mathbf{c} \in \mathbb{R}^M$ is the concentration vector for the M species included in the defined mechanism. The network therefore represents the time-dependent concentration

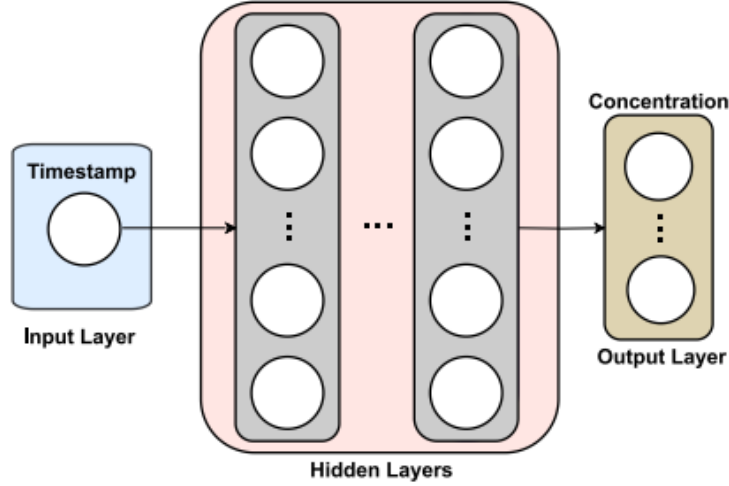


Figure 5.1: Schematic representation of a KINN. Taken from [34].

profiles of all species in the kinetic model, as illustrated in figure 5.1. The output from the forward pass is then used in the KINN loss function. The total loss is written as

$$\mathcal{L}_{KINN} = \lambda_D \mathcal{L}_D + \lambda_P \mathcal{L}_P, \quad (5.13)$$

where $\mathcal{L}_D$ is the data loss, $\mathcal{L}_P$ is the physics loss, and λ_D and λ_P are weights. The data loss is formulated as in a standard FFNN where for a datapoint n , the residual ϵ_{f_n} between the model output and the target concentration $\tilde{f}_n \in \mathbb{R}^M$ is

$$\epsilon_{f_n}(\omega_{ff}) = f(t_n, \omega_{ff}) - \tilde{f}_n, \quad (5.14)$$

The data loss is then defined as the MSE over N datapoints and M species:

$$\mathcal{L}_D(\omega_{ff}) = \frac{1}{NM} \sum_{n=1}^N \epsilon_{f_n}^T \epsilon_{f_n} \quad (5.15)$$

The defining feature of the KINN is the physics loss where the underlying physical law is the MKM given by eq. 4.11. By rearranging this expression, a physics residual ϵ_{f_n} is defined by

$$\epsilon_{f_n}(\mathbf{p}, \omega_{ff}) = \dot{\mathbf{f}}(t_n, \omega_{ff}) - \mathbf{M} \mathbf{r}_n(\mathbf{p}, \omega_{ff}) \quad (5.16)$$

for the physics loss is then defined as the MSE

$$\mathcal{L}_P(\mathbf{p}, \omega_{ff}) = \frac{1}{NM} \sum_{n=1}^N \epsilon_{f_n}^T \epsilon_{f_n} \quad (5.17)$$

where $\dot{\mathbf{f}}(t_n, \omega_{ff})$ is obtained through automatic differentiation (AD). The rate term is given by

$$\mathbf{r}_n(\mathbf{p}, \omega_{ff}) = \mathbf{k}(\mathbf{p}) \circ \psi(\mathbf{f}(t_n, \omega_{ff})) \quad (5.18)$$

where $\psi(\cdot)$ maps the concentration output to the power-law terms defined by the mechanism, and $\circ$ denotes element-wise multiplication. The rate constants are parameterized as $\mathbf{k} = \text{abs}(\mathbf{p})$ where $\mathbf{p}$ is the trainable parameter vector (also note here the isothermal assumption of $\mathbf{k}$).

Regarding the weights, λ_D and λ_P balance the interplay between the data loss and the physics loss. The data loss weight λ_D is often set to 1, while the physics loss weight λ_P is chosen by manual tuning. Finally, defining the full set of trainable parameters as $\omega_{KINN} = \{\omega_{ff}, \mathbf{p}\}$, the KINN is trained by minimizing the total loss

$$\min_{\omega_{KINN}} \mathcal{L}_{KINN}(\omega_{KINN}) \quad (5.19)$$

A gradient-based optimizer is employed to optimize the parameters until convergence, or until the specified number of training epochs is reached. In this way, the optimizer solves the forward part of the problem by fitting the concentration output and satisfying the ODEs through the optimization of ω_{ff} . The inverse problem is solved at the same time by optimizing the kinetic parameters $\mathbf{p}$ which determine the rate constants $\mathbf{k}$.

Moreover, whereas in standard PINNs the initial or boundary conditions are enforced through additional loss terms, the KINN can utilize structural enforcement of the IC through an operator $\mathbf{O}_{IC}[\cdot]$. This operator imposes the desired behavior at the initial condition t_0 , such that $\mathbf{f}(t_0, \omega_{ff}) = \mathbf{f}_0$, by using a function $\phi(t) = \tanh(t)$

$$\mathbf{O}_{IC}[\mathbf{f}(t, \omega_{ff}), \mathbf{f}_0] = \mathbf{f}(t, \omega_{ff})\phi(\kappa(t - t_0)) + \mathbf{f}_0(1 - \phi(\kappa(t - t_0))) \quad (5.20)$$

Here κ is a characteristic time parameter that controls how quickly FFNN transitions away from the imposed initial condition. It can either be chosen manually for each problem or learned within the framework. For surface species, the KINN can furthermore enforce site balance structurally in a similar manner by applying the normalization operator $\mathbf{O}_N[\cdot]$ where the output $\mathbf{f}(t, \omega_{ff})$ is mapped to the normalized output $\hat{\mathbf{f}}(t, \omega_{ff})$ through

$$\hat{\mathbf{f}}(t, \omega_{ff}) = \mathbf{O}_N[\mathbf{f}(t, \omega_{ff})] \quad (5.21)$$

for

$$\hat{f}_i = \left(1 - \sin^2(f_i)\right) \prod_{j < i} \sin^2(f_j), \quad i < M, \quad (5.22)$$

$$\hat{f}_M = \prod_{j < M} \sin^2(f_j) \quad (5.23)$$

where this trigonometric transformation directly imposes the surface site balance of eq. 4.4.

The KINN has for example successfully been adapted to TAP analysis by Nai et al. [34]. They first present a KINN model which extracts the rate constants from a TAP single-pulse, to then build a multi-pulse model which can interpolate unseen pulse behavior which is furthermore tested for a realistic scenario with noisy data.

5.3 Binarization

An underlying requirement of the functions used inside a FFNN can be understood from the backpropagation algorithm. Since the network parameters are updated using gradients, the operations in the network should be differentiable. This becomes a problem if one wants to use a hard binarization function inside the network, for example to force a neuron or parameter to become either 0 or 1. A binary step function can be written as

$$b(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad (5.24)$$

This type of function is useful if a binary decision is needed, but it is problematic during training. Standard backpropagation relies on gradients to update the model weights. If the gradient is zero or undefined, the update cannot flow through the binarization operation to the earlier layers. For the binary step function, the derivative is zero almost everywhere and undefined at the jump point. If this true derivative is used, the gradient becomes zero almost everywhere, which prevents the weights from being updated.[54]

This can be seen from the chain rule in backpropagation. Looking at for example the middle derivative $\frac{\partial a}{\partial z}$ in eq. 5.8, one can see that if the activation function is replaced by a hard binarization function, this derivative becomes problematic. To circumvent this, one possible solution is to use a straight-through estimator straight-through estimator (STE) [1]. The idea of the STE is to use the hard binary function in the forward pass, while approximating its derivative during the backward pass. For a binary activation $a = b(z)$ the backward derivative can be approximated as

$$\frac{\partial a}{\partial z} \approx 1 \quad (5.25)$$

This means that the backward pass treats the binarization function as if it was the identity function. The gradient calculated for the output of the binarization node is therefore passed straight through to the input of the binarization node. In backpropagation, eq. 5.8 would then be approximated as

$$\frac{\partial \mathcal{L}}{\partial W_{11}^{(1)}} \approx \left(\frac{\partial \mathcal{L}}{\partial a_1^{(2)}} \right) \cdot 1 \cdot \left(\frac{\partial z_1^{(2)}}{\partial W_{11}^{(1)}} \right) \quad (5.26)$$

The STE is therefore not the true derivative of the binarization function, but just an approximation that allows the model to be trained with gradient based optimization. The identity approximation is the simplest and most common form of the STE, but multiple versions also exists. [54]

A simpler way of obtaining binary values in ML is to formulate the problem as binary classification.[18, 26] Here the model produces a continuous output between 0 and 1 using a sigmoid activation function

$$p = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (5.27)$$

Here p can be interpreted as the predicted probability of belonging to class 1. Since the sigmoid function is differentiable, the model can be trained with standard backpropagation without needing an approximation such as the STE. The continuous output can then be converted to a hard binary value by applying a threshold, usually

$$\hat{y} = \begin{cases} 1 & p \geq 0.5 \\ 0 & p < 0.5 \end{cases} \quad (5.28)$$

Part III

Formulating the model - Methodology

Chapter 6

M-KINN

The aim of this project has been to develop a ML model to aid TAP data analysis. Whereas previous work on this topic by Nai et al. [34] has successfully extracted rate constants from single-pulse data using KINNs [17], the goal of this effort was to work toward realistic multi-pulse model, thus utilizing the single-pulse model more as a stepping stone. Therefore we realized the potential of improvement for a single-pulse model. Particularly, to train the KINN, one must assume a given mechanism to test for and continue manually constructing mechanisms until convergence. This would in practice mean having to construct and train multiple KINNs. Hence, the model we work toward is centered around the advantage of being able to test multiple mechanisms at the same time, as well as potentially inferring mechanisms not identified. The value of such a model furthermore becomes increasingly more significant for reactions which possibly have a large number of elementary reactions and intermediate species such as combustion reactions.[56] In this project we propose M-KINN, a PyTorch [21] based ML model for TAP which utilizes an inverse KINN with additional custom loss term to simultaneously fit both rate constants and mechanism from data. A trained model returns a proposed mechanism alongside information about the kinetics of each of the elementary reactions.

Central to M-KINN is a **supermechanism** which contains all the possible elementary steps to search in. This is in practice a text file where the contents can either be user-defined or automatically generated by a automatic chemical reaction mechanism generator such as RMG [23].

mechanismfile.txt

```
C0 + * <-> C0*  
O2 + 2* -> 2O*  
C0* + O* <-> C02 + 2*  
C0 + O* -> C02 + *
```

For each of the listed reactions, a rate constant and reaction switch (which indicates if a reaction is active or not) is defined and optimized. The model is furthermore defined with a simple "interface" which takes this supermechanism file alongside additional files, such as a data file and optionally a hyperparameter configuration file, and prints the list of proposed elementary reactions, rate constants and free energy of activation $\Delta G^\ddagger$ together with the concentration profiles.

```
>>> python train.py -n run_name -m mechanism_file.txt
    ↪ -d data_file.txt
```

```
Reaction mechanism:
-----
r1: .... | k = ... | delta_G = ...
r3: .... | k = ... | delta_G = ...
-----
```

Like the TAP-KINN work of Nai et al., this model is built starting from an "ideal scenario" where reactor information from the catalyst zone, such as concentration and net flux, is available from the simulation. This in contrast with a practical scenario, which mirrors a realistic case where only fluxes at the reactor outlet are assumed to be known. This practical scenario represents the next natural step for the model. In this case, one would hope to take advantage of the general ability of PINNs to address problems described by limited data or noisy experimental observations.

6.1 FFNN Architecture

A KINN $\mathbf{f}(t, \omega_{ff})$ as described in section 5.2 is utilized with the modification that time t is mapped to concentration $\mathbf{f}_g(t, \omega_{ff}) \in \mathbb{R}^{M_g}$ for the M_g gas species and fractional coverage $\mathbf{f}_s(t, \omega_{ff}) \in \mathbb{R}^{M_s}$ for the M_s surface species.

$$\mathbf{f}(t, \omega_{ff}) = \begin{bmatrix} \mathbf{f}_g(t, \omega_{ff}) \\ \mathbf{f}_s(t, \omega_{ff}) \end{bmatrix} \quad (6.1)$$

The structural enforcement of initial conditions $\mathbf{O}_{IC}[\cdot]$ in the KINN, as described by eq. 5.20, is treated as optional. The original TAP-KINN model does not enforce initial conditions as the multi-pulse case it builds towards it may not be known and it is shown that the model is able to perform well despite the exclusion. In this work, both cases are therefore tested. When initial conditions are enforced, the gas species are initialized to zero, while the surface species are initialized according to the assumed catalyst state. An overview of values is given in the data section.

Furthermore, because the surface species are represented as fractional coverages, the site balance conservation must also be considered. Two approaches are tested in this work. In the first approach, the site balance is enforced structurally using the normalization operator $\mathbf{O}_N[\cdot]$ as described in section 5.2. In this case, the raw surface output is mapped to normalized coverages before being used in the kinetic model. The second approach does not enforce normalization directly in the architecture. Instead, the site balance is included as an additional loss term as described in the discussion of the loss function below.

6.2 Custom loss function

The loss function of M-KINN is given by

$$\mathcal{L} = \lambda_D \mathcal{L}_D + \lambda_P \mathcal{L}_P + \lambda_{SB} \mathcal{L}_{SB} + \lambda_{MC} \mathcal{L}_{MC} + \lambda_{SR} \mathcal{L}_{SR} \quad (6.2)$$

where the standard data and physics loss terms $\mathcal{L}_D$ and $\mathcal{L}_P$ are extended with three custom loss terms. The model introduces a site balance term $\mathcal{L}_{SB}$, a mechanistic consistency term $\mathcal{L}_{MC}$, and a switch regularization term $\mathcal{L}_{SR}$. Additionally, the physics loss is extended as the central part of the model.

The data loss, given by Eq. 5.14 and Eq. 5.15, is redefined to be calculated over the M_a species for which data is available,

$$\mathcal{L}_D(\omega_{ff}) = \frac{1}{NM_a} \sum_{n=1}^N \epsilon_{f_n}^T \epsilon_{f_n} \quad (6.3)$$

where $\epsilon_{f_n} \in \mathbb{R}^{M_a}$ is the residual between the model output and the available target data which corresponds to concentrations and coverages in the active zone of the TAP reactor. The physics loss for the gas species follows from the TAP active zone balance in Eq. 4.14. Through approaching a thin zone approximation, the diffusion term is written as the transport contribution g_n giving the residual

$$\epsilon_{f_{g,n}}(\mathbf{p}_s, \mathbf{p}_k, \omega_{ff}) = \epsilon_b \dot{\mathbf{f}}_g(t_n, \omega_{ff}) - (g_n - (1 - \epsilon_b) a_v C_{s,*,tot} \mathbf{M}_g \mathbf{r}_n(\mathbf{p}_s, \mathbf{p}_k, \omega_{ff})) \quad (6.4)$$

where ϵ_b is the bed void fraction, a_v is the specific surface area, $C_{s,*,tot}$ is the total concentration of active sites, and $\mathbf{M}_g$ is the stoichiometric matrix defined for the gas species. The transport term is approximated from the net flux through the catalyst zone,

$$g_{A,n} = \frac{f_{A,n}^{in} - f_{A,n}^{out}}{L_{cat}} \quad (6.5)$$

where $f_{A,n}^{in}$ and $f_{A,n}^{out}$ denote the flux into and out of the active zone for species A , and L_{cat} is the length of the catalyst zone.

For the surface species there is no transport term and residual follows from the TAP surface balance in Eq. 4.15

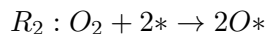
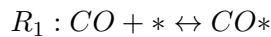
$$\epsilon_{f_{s,n}}(\mathbf{p}_s, \mathbf{p}_k, \omega_{ff}) = \dot{\mathbf{f}}_s(t_n, \omega_{ff}) - \mathbf{M}_s \mathbf{r}_n(\mathbf{p}_s, \mathbf{p}_k, \omega_{ff}) \quad (6.6)$$

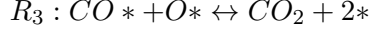
The physics loss $\mathcal{L}_P$ is then calculated from the gas and surface residuals in the same MSE form as in Eq. 5.17.

Beyond this the essential extension to the physics loss is the possibility of selecting elementary steps from the supermechanism. This is encoded in the rate term

$$\mathbf{r}_n(\mathbf{p}_s, \mathbf{p}_k, \omega_{ff}) = \mathbf{s}(\mathbf{p}_s) \odot (\mathbf{k}(\mathbf{p}_k) \circ \psi(\mathbf{f}(t_n, \omega_{ff}))) \quad (6.7)$$

where $\mathbf{p}_s \in \mathbb{R}^R$ is the trainable reaction switch parameter vector for the R elementary reactions in the supermechanism. The reaction switches are defined as $\mathbf{s} = \mathbf{h}(\mathbf{p}_s)$, where $\mathbf{h}$ maps the trainable parameters to the desired switch range. A switch value $s_i = 0$ corresponds to reaction i not participating, while $s_i = 1$ indicates that the elementary reaction is active. The options for the binary reaction switch scheme $\mathbf{h}(\cdot)$ is laid out in section 5.3 as either a STE like approach or with the use of a sigmoid activation function. The kinetic parameters are furthermore constrained to be non-negative by defining $\mathbf{k} = \text{abs}(\mathbf{p}_k)$ where $\mathbf{p}_k$ is the trainable kinetic parameter vector as in the original KINN formulation. To illustrate this, a supermechanism defined as





would for a given datapoint give the following contribution to the rate of $O*$,

$$R_{O*} = 2s_2k_2C_{O_2}\theta_*^2 - s_3k_3\theta_{CO*}\theta_{O*} + s_{-3}k_{-3}C_{CO_2}\theta_*^2$$

If then only dissociative chemisorption occurred, minimizing the physics loss should drive the model toward $s_1 = 0$, $s_2 = 1$, and $s_3 = 0$.

As for the three additional custom loss terms, a sorting of two subcategories can be made based on the nature of the terms. The first subcategory enforces additional physical correctness. One such loss term is $\mathcal{L}_{SB}$ which enforces conservation of the fractional coverages to unity

$$\mathcal{L}_{SB}(\omega_{ff}) = \frac{1}{N} \sum_{n=1}^N \left(1 - \sum_{j=1}^{M_s} f_{s,j}(t_n, \omega_{ff}) \right)^2 \quad (6.8)$$

This is an alternative to the structurally enforced normalization scheme discussed in Section 5.2. If structural normalization is used, the surface coverages are already constrained to sum to unity through the model output and the loss term can be neglected.

The other physical correctness term is the mechanistic consistency term $\mathcal{L}_{MC}$. Enforcing a consistent mechanism means that the model should not have an active elementary reaction which requires a reactant that is not available. A LH-like reaction step between two adsorbed species should for example be discouraged from activating without also activating steps which produce these adsorbed species. Here, $\mathbf{Q} \in \mathbb{R}^{M \times R}$ denotes the matrix for the required species, with elements Q_{ij} defined as

$$Q_{ij} = \begin{cases} 1, & \text{if species } i \text{ is a reactant in reaction } j \\ 0, & \text{otherwise} \end{cases} \quad (6.9)$$

Similarly, $\mathbf{Z} \in \mathbb{R}^{M \times R}$ denotes the matrix for the supplied species, with elements Z_{ij} defined as

$$Z_{ij} = \begin{cases} 1, & \text{if species } i \text{ is a product in reaction } j \\ 0, & \text{otherwise} \end{cases} \quad (6.10)$$

The externally available species, meaning the pulsed gas reactants and free surface sites, are encoded in $\mathbf{b} \in \mathbb{R}^M$. Species that the mechanism must explain, such as observed products, are encoded in $\mathbf{q} \in \mathbb{R}^M$. The consistency error is then

$$\mathbf{e} = \min(\mathbf{1}, \mathbf{Q}\mathbf{s} + \mathbf{q}) - \min(\mathbf{1}, \mathbf{b} + \mathbf{Z}\mathbf{s}) \quad (6.11)$$

where the minimum is applied element wise. The loss is

$$\mathcal{L}_{MC}(\mathbf{p}_s) = \sum_{i=1}^M \text{ReLU}(e_i) \quad (6.12)$$

where ReLU only penalizes positive values thus allowing possible valid spectator species. Lastly, the final subcategory and custom loss term $\mathcal{L}_{SR}$ enforces regularization of the reaction switches so that the model gets an incentive to only activate the minimum required reactions in the supermechanism

$$\mathcal{L}_{SR}(\mathbf{p}_s) = \sum_{i=1}^R s_i \quad (6.13)$$

Regarding the loss weights, a magnitude-based approach presented by Zhang et al. [56] is used. In short, the magnitudes of the individual loss terms are estimated and used to scale the terms to a similar range. Some loss terms are further relaxed so that the training mainly focuses on the data and physics losses. The additional loss terms here will similarly be relaxed. It should also be noted that the magnitudes of the mechanistic consistency and switch regularization terms can be estimated without performing any training runs as they at a maximum can equal to the amount of species and amount of reactions respectively.

Finally, the total amount of trainable parameters for M-KINN is defined as

$$\omega = \{p_s, p_k, \omega_{ff}\} \quad (6.14)$$

giving the optimization problem

$$\begin{aligned} \min_{\omega} \mathcal{L}(\omega) = & \lambda_D \mathcal{L}_D(\omega_{ff}) + \lambda_P \mathcal{L}_P(p_s, p_k, \omega_{ff}) \\ & + \lambda_{SB} \mathcal{L}_{SB}(\omega_{ff}) + \lambda_{MC} \mathcal{L}_{MC}(p_s) + \lambda_{SR} \mathcal{L}_{SR}(p_s) \end{aligned} \quad (6.15)$$

An overview of the model can be seen in Fig. 6.1. After training the optimized reaction

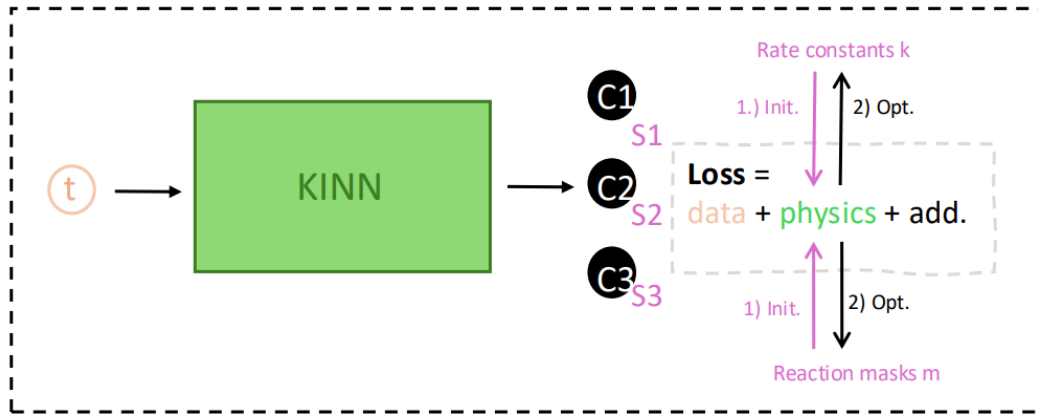


Figure 6.1: Schematic representation of M-KINN

switches are used to infer the active mechanism from the supermechanism. The fitted kinetic parameters are then used to calculate the corresponding free energies of activation through the Eyring equation in Eq. 4.7.

Chapter 7

Data

In the ideal case, where the goal is to test the ability of the model to recover the inverse parameters, noise-free synthetic data from TAPSolver [55] is used. The data contains active thin zone concentrations and flux values. As TAPSolver defines the gas active zone PDE without the prefactor terms in front of the rate term, the gas species physics residual is therefore changed to

$$\epsilon_{\dot{f}_{g,n}}(\mathbf{p}_s, \mathbf{p}_k, \boldsymbol{\omega}_{ff}) = \epsilon_b \dot{f}_g(t_n, \boldsymbol{\omega}_{ff}) - (\mathbf{g}_n - \mathbf{M}_g \mathbf{r}_n(\mathbf{p}_s, \mathbf{p}_k, \boldsymbol{\omega}_{ff})) \quad (7.1)$$

to match the data.

TAPSolver furthermore gives the surface species as surface concentrations. Since the MKM in this work is formulated in terms of fractional coverages, the surface concentrations are first converted using

$$\theta_i = \frac{C_{s,i}}{C_{s,*,tot}} \quad (7.2)$$

where $C_{s,i}$ is the surface concentration of species i , and $C_{s,*,tot}$ is the total concentration of active surface sites chosen in the data generation.

7.1 Data preparation

Before training, the data is prepared to make the model easier to train and to improve resolution in the most important parts of the transient response. This follows the general approach used in the TAP-KINN work.

1. **Non-uniform sampling:** The data is sampled with more points in regions where the concentration profiles change quickly. This gives better resolution in high-derivative regions while keeping the number of training points lower. The sequence is chosen so that the sample sizes are multiples of 2, 5, 8, 10, and 20.
2. **Logarithmic time transformation:** Instead of using the raw time values t , the time is transformed using a natural logarithm. This gives better resolution at short times where the concentration profiles often change fastest.
3. **Normalization:** Normalization is used to make training more stable and efficient. When normalization is not built directly into the model through the structural enforcement, the label data is normalized instead. The surface species are represented as fractional coverages, while the gas-phase species are normalized using max scaling where each species is divided by its largest value.

The logarithmic time transformation creates one issue when initial conditions are enforced structurally. Since $\log(0)$ is not defined, the initial time point cannot be used directly. This was not an issue in the original TAP-KINN model as the initial conditions are not enforced. In this work, initial conditions are tested as an optional model feature, and so when they are enforced, a small timestep shift is added before applying the logarithmic transformation shifting all datapoints in the mapping.

7.2 Reaction system

The TAP-KINN model tests the single-pulse case using CO oxidation data generated from a combined LH and ER mechanism. To test whether the mechanism can be recovered from the data, this work expands the reaction set with additional candidate reactions. The model is evaluated on a smaller supermechanism containing nine elementary reactions. Six of these correspond to the true mechanism and the remaining three reactions are inactive candidate reactions. The inactive reactions are included only as candidate reactions in the supermechanism. They are not used to generate the data. A successful model should therefore assign low switch values to these reactions.

Table 7.1: Elementary reactions included in the supermechanism

Elementary reaction	Rate constant
$CO + * \rightarrow CO*$	$k_1 = 15.0$
$CO* \rightarrow CO + *$	$k_2 = 0.70$
$O_2 + 2* \rightarrow 2O*$	$k_3 = 0.01$
$CO* + O* \rightarrow CO_2 + 2*$	$k_4 = 0.40$
$CO_2 + 2* \rightarrow CO* + O*$	$k_5 = 0.0006$
$CO + O* \rightarrow CO_2 + *$	$k_6 = 15.2$
$CO_2* \rightarrow CO_2 + *$	$k_7 = -$
$CO + O* \rightarrow CO_2*$	$k_8 = -$
$O_2 + * \rightarrow O_2*$	$k_9 = -$

Note that the rate constants given here differ from the values in the original TAP-KINN work for the reactions involving more than 2 active sites. This is due to that the MKM in this thesis is formulated in terms of fractional coverages.

For the initial conditions, the gas species are set to zero. This is consistent with the evacuated TAP reactor. The adsorbed species are also initialized to zero, while the vacant site coverage for the catalyst is initialized to one. This corresponds to a clean catalyst surface where all active sites are initially free. These values are only enforced directly in runs where the structural initial-condition operator $\mathbf{O}_{IC}[\cdot]$ is used. In runs without structural initial conditions, the model is trained without explicitly imposing the initial state.

Part IV

Evaluating the model - Results and discussion

Chapter 8

M-KINN

Serving as a starting point, a reference model with giving settings of the model is presented and evaluated in the first section. The better choice model is then carried onto and developed further in the following section where the evaluation of two alternative approaches to the reference model is explored.

8.1 Reference model

The reference model is made up of the following model settings

1. Switch scheme: STE
2. Normalization scheme: loss term.

For the first run of this model, ICs are not enforced. The run detailed run settings are shown in table 8.1.

Table 8.1: Settings of the first reference run.

Run overview	
Setting	Option
Switch scheme	STE
Normalization scheme	loss term
IC	excluded
Weights	
Weight	Value
λ_D	1
λ_P	50
λ_{SB}	0.002
λ_{MC}	0.01
λ_{SR}	0.00001

Firstly, to illustrate an example for the calculation of the weights, the magnitude of the data, physics and SB term were chosen to be 0.1, 0.002 and 0.002 respectively. Furthermore, with 9 reactions and 8 species, the SR and MC lie in this range. This leads to the weights shown in table using the data loss magnitude to scale while relaxing the SB and MC term by a factor of 10^{-2} and SR with 10^{-4} as the system is relatively small.

The training curves for the resulting run are shown in figure 8.1.

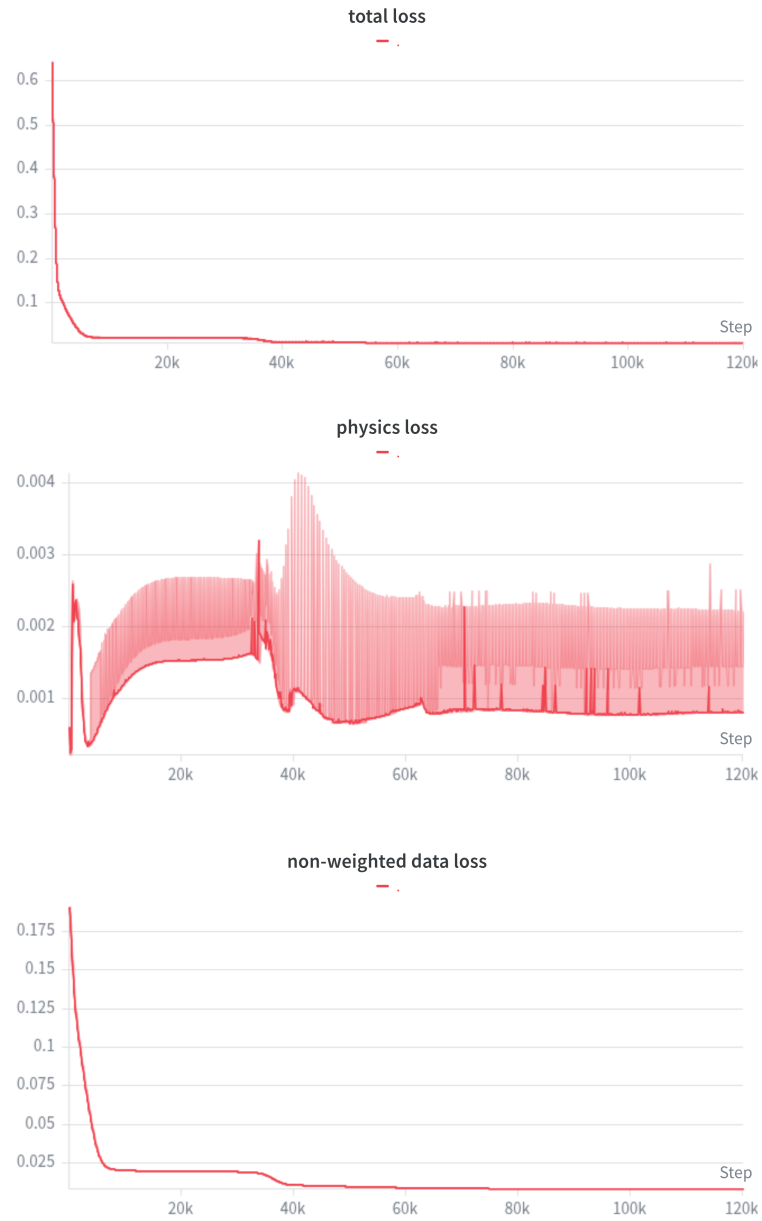


Figure 8.1: Total, physics and data loss curves for training of the first reference model.

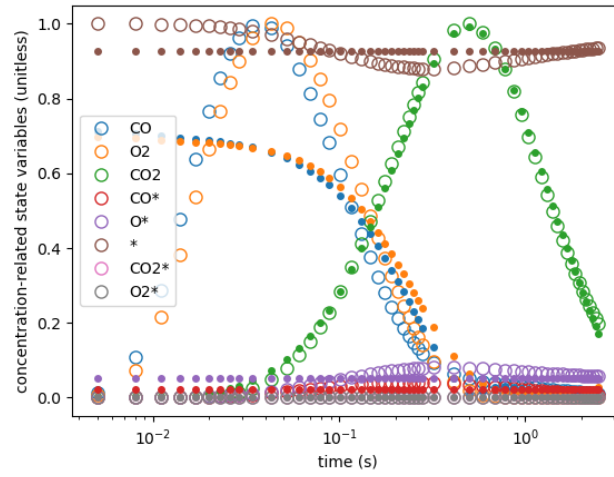


Figure 8.2: SB, MC and SR loss curves for training of the first reference model.

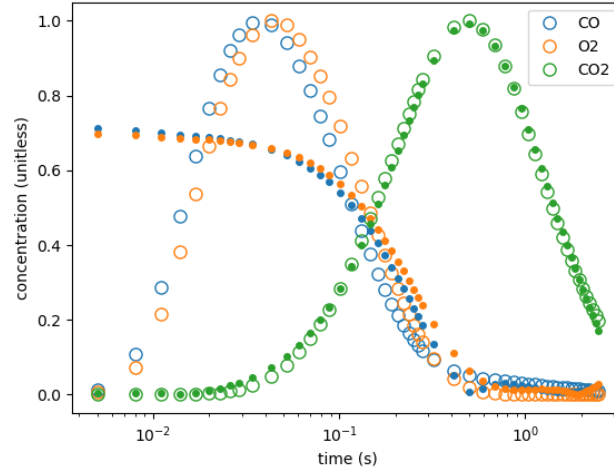
The total loss shows an overall smooth convergence to error values on the order of magnitude of 10^{-2} . The very steep decrease in loss is likely a result from the random initialization of the neural network, leading to large errors in the first epoch. The data loss exhibits a similar shape, indicating that it contributes to the steep decrease at the beginning of training. Similarly the SB penalty loss has a similar shape which means that the model quickly learns that the surface species should sum up to one. The physics loss shows an oscillating behavior, where the loss decreases overall but occasionally increases again before continuing to improve. This is a common effect in physics-informed neural networks, as the model must simultaneously satisfy the physical equations as well as the boundary conditions and training data. Temporary dips to very low loss values, also observed here, can occur when the network finds a solution that

matches the physics particularly well in one part of the problem, but later adjusts to better balance all training objectives. Despite these fluctuations, the overall downward trend and final stabilization of the loss indicate that the model is learning a physically meaningful solution." [39]

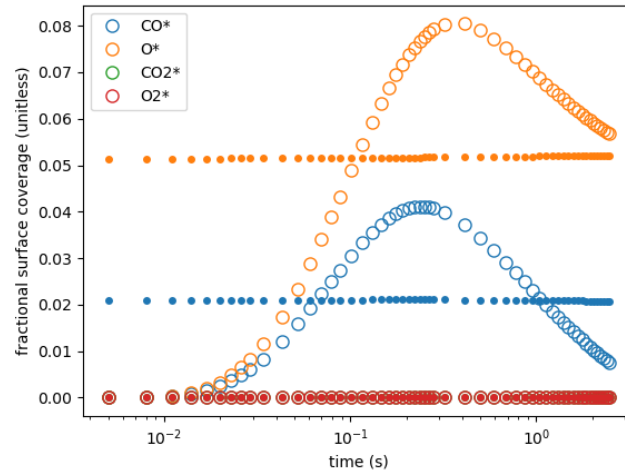
The concentration profiles produced by this run are shown in figure 8.3



(a) All species



(b) Gas species



(c) Surface species (except *)

Figure 8.3: Concentration curves of the first reference run.

From the concentration curves of the gas species it can be observed that the model accurately reproduces the CO_2 values but fails to capture the progression of CO and

O₂. The assumed cause for this was, that this run was missing the inclusion of proper ICs such that the concentrations curves fail to match the true values especially in the beginning. Similarly, for the surface species CO₂* and O₂* are captured fairly accurately, while the model fails for CO* and O*. In general, the model seems to be unable to properly predict changes in the fractional surfaces coverages of adsorped species. Overall, this run produced an MSE of 0.008 with an R^2 value of 0.56, highlighting poor agreement with the data.

Table 8.2 shows the predicted and true values for the reaction switches and rate constants. From these results, it is obvious that the model was not able to properly capture which reactions switches should be on or off with four wrongly predicted to be off and one wrongly predicted to be on. Furthermore, the predicted rate constants also do not match the ground truth values with partially large deviations.

Table 8.2: True and predicted switches and rate constants of the first reference model.

Reaction	1	2	3	4	5	6	7	8	9
s_{opt}	0	0	1	0	0	1	0	1	0
s	1	1	1	1	1	1	0	0	0
k_{opt}	0.003	0.0001	0.04	0.0009	0.005	0.9	0.007	0.4	0.00001
k	15.0	0.70	0.01	0.40	0.0006	15.2	-	-	-

The incorrect fit of the earlier timepoints of the concentration curves in the first reference model motivated the enforcement of ICs. More precisely, the idea was to include ICs in order to condition the model to find solutions that create gas concentration profiles that agree with the ground truth values in the early timepoints. A run that enforces IC was set up with $\kappa = 0.02$ and a time step size of 0.001 and detailed settings listed in table 8.3.

Table 8.3: Settings of the second reference run.

Run overview	
Setting	Option
Switch scheme	STE
Normalization scheme	loss term
IC	included
Weights	
Weight	Value
λ_D	1
λ_P	10
λ_{SB}	0.5
λ_{MC}	0.0005
λ_{SR}	0.0000006

The corresponding loss curves are shown in figures 8.4 and 8.5.

The overall shape of loss curves are very similar to the ones obtained for the first reference model. However, the data loss exhibits a less steep convergence towards similar final loss values. Whereas in the first reference model the data loss plateaued after 20k epochs, the data loss for reference model shows a fairly steady decrease before converging to much lower error values. This indicates that the second reference model could find a

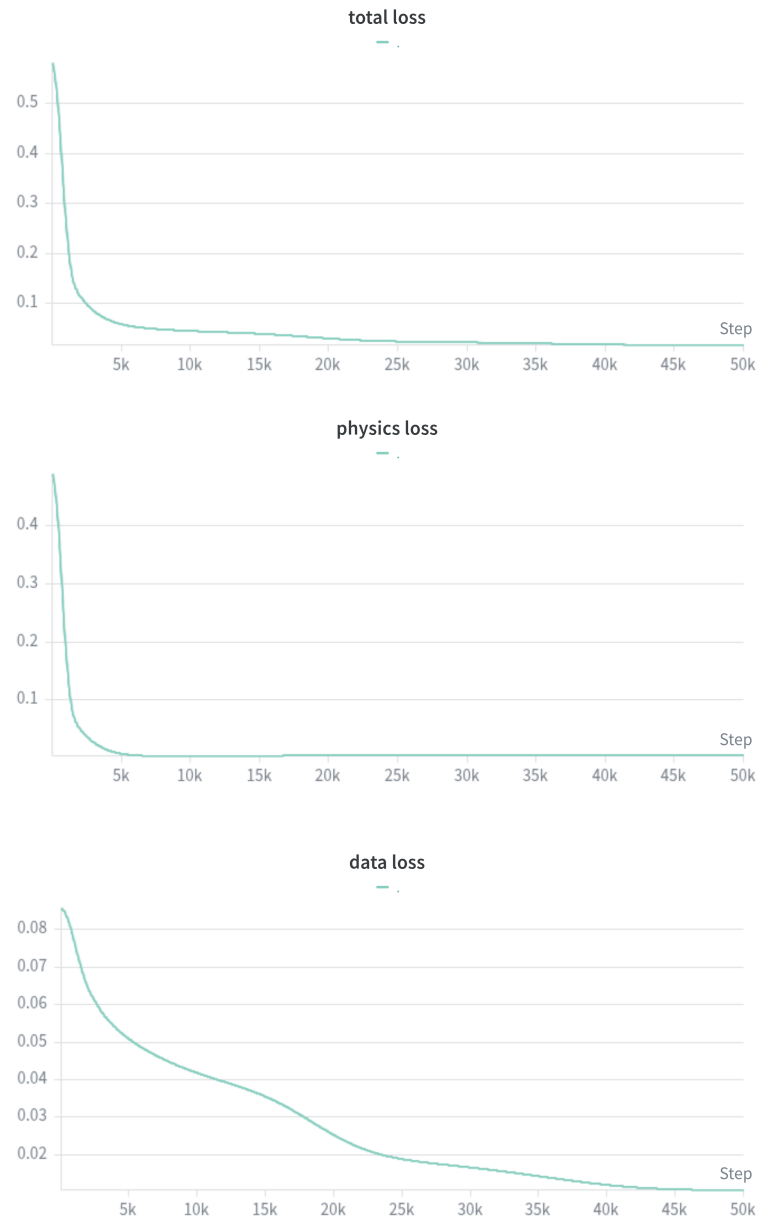


Figure 8.4: Total, physics and data loss curves for training of the second reference model.

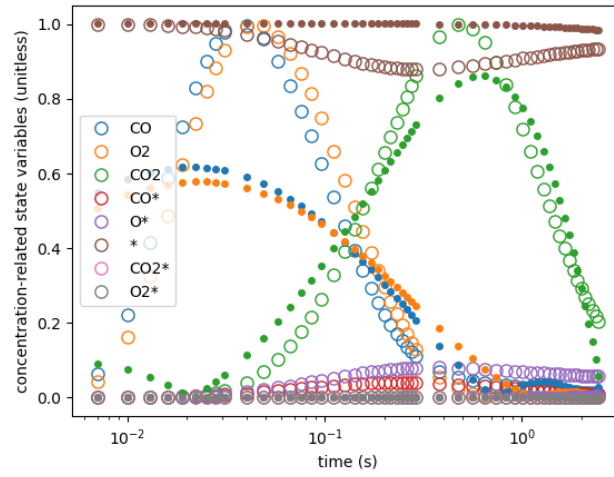


Figure 8.5: SB, MC and SR loss curves for training of the second reference model.

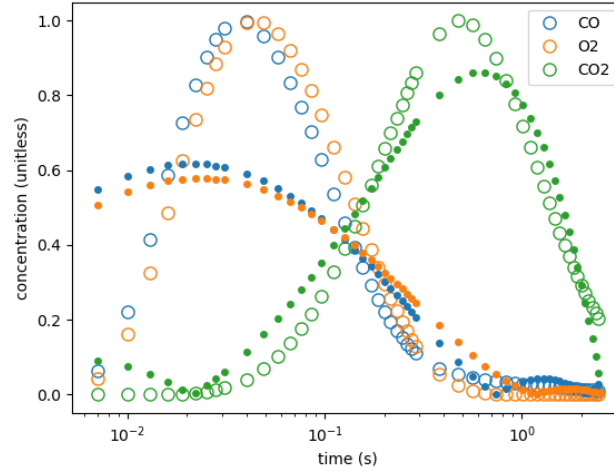
weight configuration that is able to reproduce the given data much better. Furthermore, the physics loss reveals a significantly smoother convergence behavior. This observation may indicate that the inclusion of the initial condition stabilizes the optimization process and renders the underlying inverse problem less ill-posed. By constraining the solution space more strongly at the early timepoints, the model may be guided toward more consistent parameter regions, even if this does not necessarily translate into improved agreement with the full experimental trajectories (see below). The curves for the switch regularization and mechanism consistency losses are very similar between the first and second reference model. Overall, convergence is much quicker compared to the first reference model, with only 50k instead of 200k epochs.

An interesting detail is that while in the first reference model the physics loss starts at already low values whereas the data loss starts high and converges, it is the exact opposite in the second reference model. This behaviour can be explained by the fact that when not enforcing the initial condition, the model has more freedom in shaping the concentration trajectories and can therefore more easily find solutions that satisfy the governing differential equations, leading to a very small initial physics loss. However, these solutions do not necessarily agree well with the data, resulting in a comparatively large data loss at the beginning of training. In contrast, by including the initial condition the solution space is more strongly constrained from the start, making it more difficult for the model to satisfy the physical equations everywhere in the time domain, leading to a larger initial physics loss.

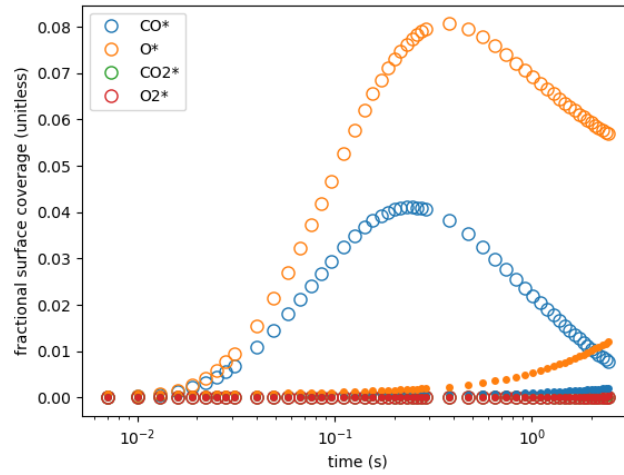
The concentration profiles produced by the second reference run are shown in figure 8.6



(a) All species



(b) Gas species



(c) Surface species (except *)

Figure 8.6: Concentration curves for reference run with IC.

The obtained concentration profiles are similar to those of the first reference run in that the progression of gas CO_2 is reproduced fairly well. However, it should be noted that

the first reference model has better agreement with the data. The shape of CO and O₂ fits better with a clear upwards arc before diminishing towards the later timepoints, whereas in the first reference model their concentrations plateaued from the beginning. Investigation of the model output at the initial condition reveals that its inclusion enables the model to accurately reproduce the corresponding experimental values. The profiles for the adsorbed species are also very similar with the only change that the O* species now shows an upward trend at the end of the time domain. Table 8.4 shows the predicted and true values for the reaction switches and rate constants. Similar to the first reference model, the second reference model was not able to reproduce the correct reaction switches with only two correctly predicted to be on. However, all switches that were supposed to be off, were also correctly predicted to be off. The obtained rate constants however, differ significantly from the ground truth values.

Table 8.4: True and predicted switches and rate constants of the second reference model.

Reaction	1	2	3	4	5	6	7	8	9
s_{opt}	0	0	0	0	1	1	0	0	0
s	1	1	1	1	1	1	0	0	0
k_{opt}	0.0004	0.00006	0.0002	0.1	0.000007	2.3	0.002	0.00006	0.0001
k	15.0	0.70	0.01	0.40	0.0006	15.2	-	-	-

Overall the second reference model performs similarly to the first reference model with an MSE of 0.011 and corresponding R^2 value of 0.54. As the two runs have a qualitative similar fit and the IC contribution enables smoother and quicker convergence behavior, the IC enforcement has been included in the Investigation of further models.

8.2 Beyond the reference model

8.2.1 Alternative reaction mask scheme

To investigate whether the a different switching scheme could alleviate some of the problems encountered before, we experimented with the sigmoid scheme with the detailed settings shown in table 8.5.

Table 8.5: Settings of the run with sigmoid switching scheme.

Run overview	
Setting	Option
Switch scheme	sigmoid
Normalization scheme	loss term
IC	included
Weights	
Weight	Value
λ_D	1
λ_P	10
λ_{SB}	0.25
λ_{MC}	0.000055
λ_{SR}	0.00000062

The corresponding loss curves are shown in figures 8.7 and 8.8.

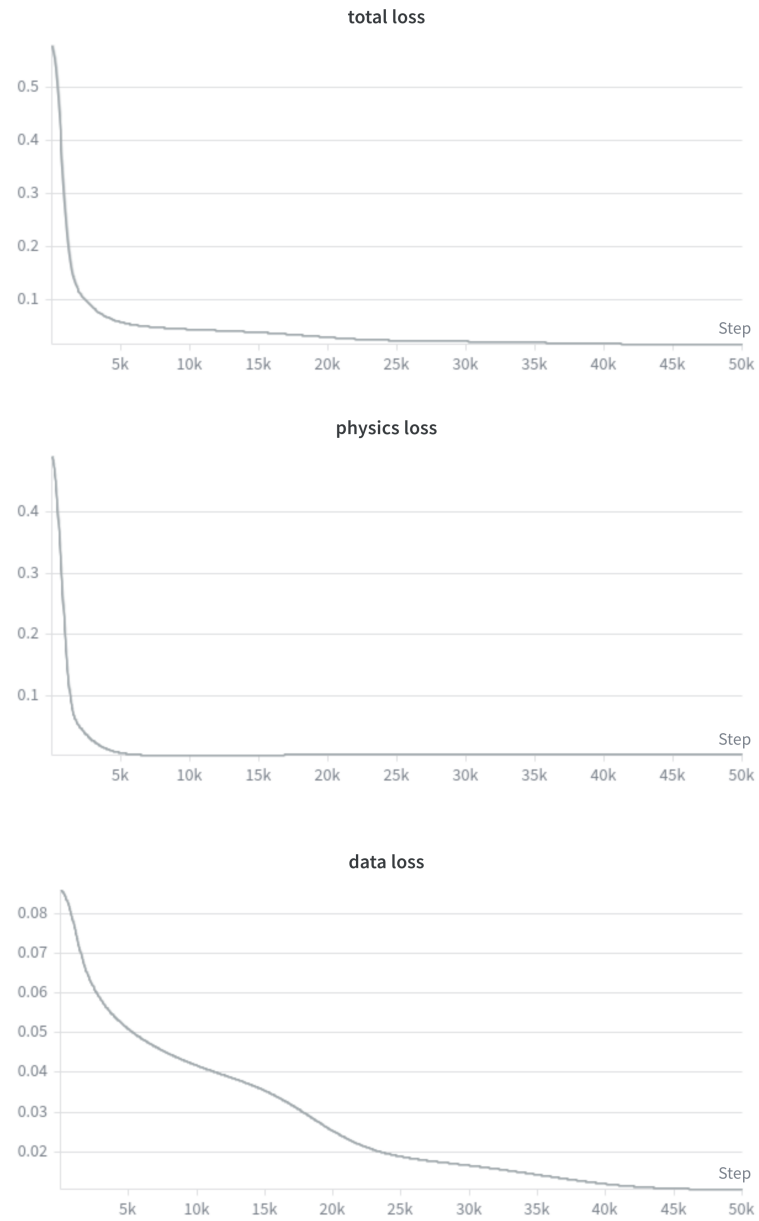


Figure 8.7: Total, physics and data loss curves for training of the the model with a sigmoid switching function.

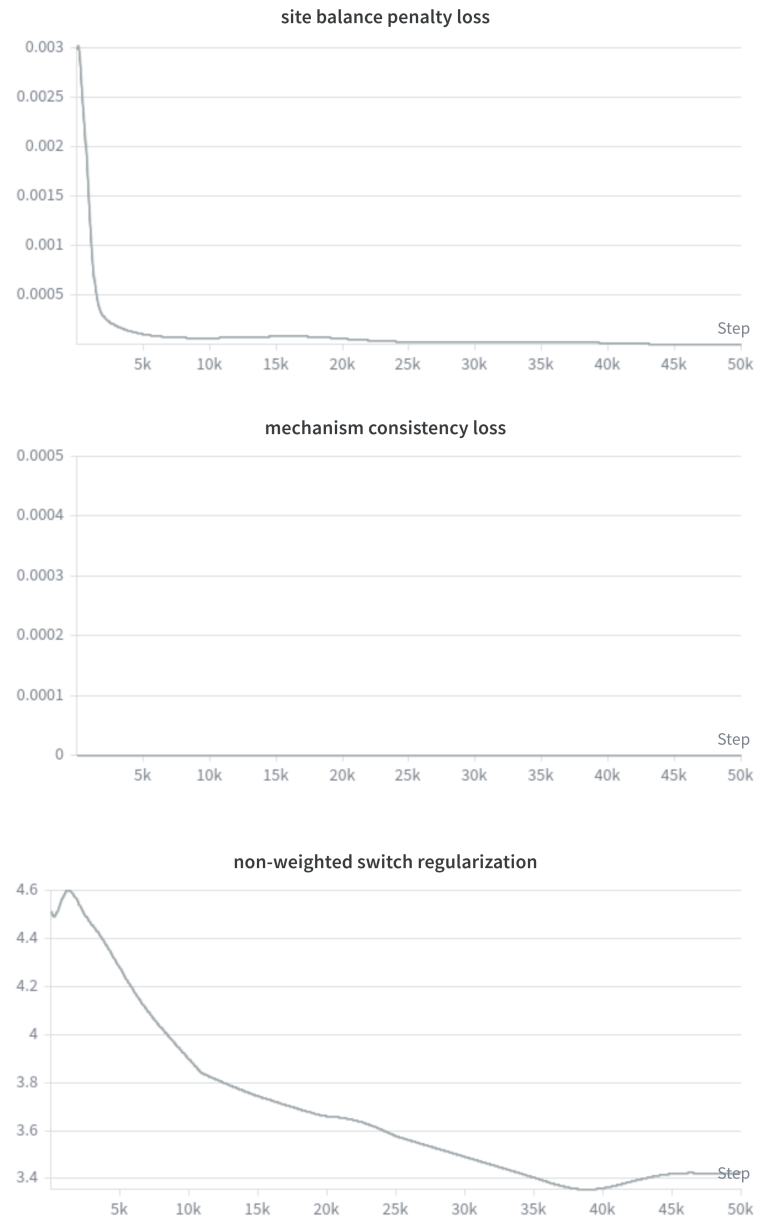
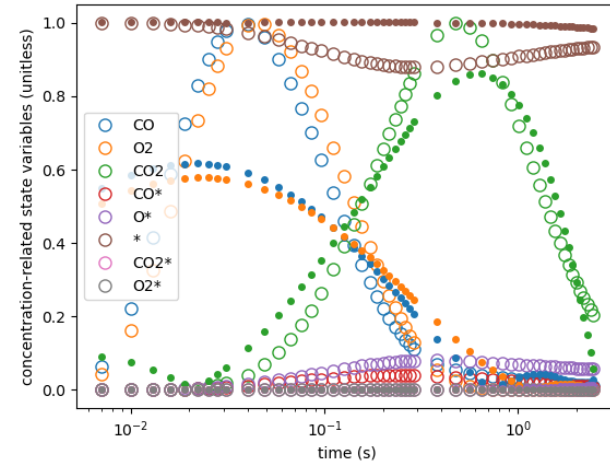


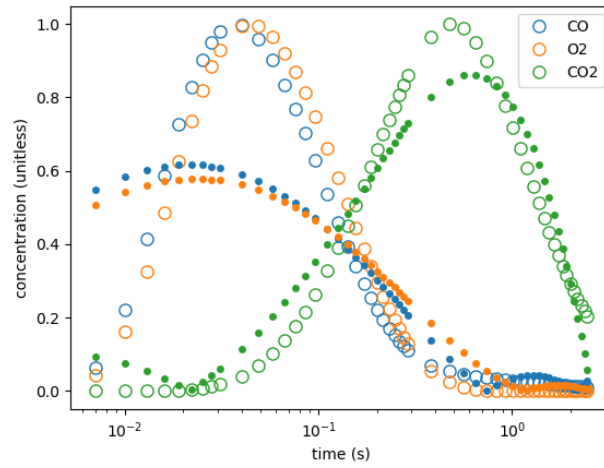
Figure 8.8: SB, MC and SR loss curves for training of the model with a sigmoid switching function.

Overall, the same trends as in the second reference model can be observed with generally smooth convergence with a dominating physics loss and lower data loss as discussed above. A key difference can be seen for the switch regularization loss though, where in this run its convergence behavior is much smoother compared to the more erratic and step-wise shapes obtained with both reference models. This result reflects the fact that the sigmoid function allows the model to smoothly tune the reaction switches whereas the STE approach causes a binarization that leads to the observed step-like behavior.

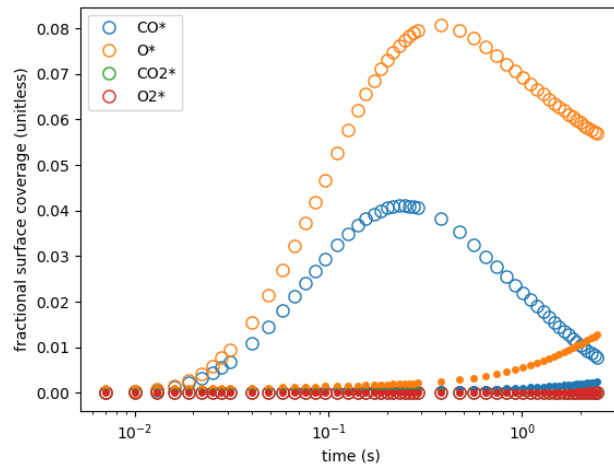
The concentration profiles produced by this run are shown in figure 8.9



(a) All species



(b) Gas species



(c) Surface species (except *)

Figure 8.9: Concentration curves for the model with a sigmoid switching function.

Compared to the second reference run there are no significant qualitative difference in the concentration profiles of any species. This indicates that the change of the switching

function has not improved the overall accuracy of the model as supported by an obtained MSE of 0.011 and R^2 value of 0.54. The predicted switches and rate constant in table ?? are also similar to those of the second reference model. Note that due to the sigmoid function, the switches are float valued between 0 and 1 where anything above 0.5 can be considered "on". Again, while the switches of the three fake reaction mechanisms is correctly set to 0, the model only correctly turns two of the six true mechanisms on.

Table 8.6: True and predicted switches and rate constants of the model with a sigmoid switching function.

Reaction	1	2	3	4	5	6	7	8	9
s_{opt}	0.37	0.043	0.49	0.24	0.51	0.65	0.4	0.4	0.33
s	1	1	1	1	1	1	0	0	0
k_{opt}	$2 \cdot 10^{-6}$	0.0001	$4 \cdot 10^{-6}$	10^{-7}	$5 \cdot 10^{-7}$	2.8	0.07	$3 \cdot 10^{-7}$	$2 \cdot 10^{-6}$
k	15.0	0.70	0.01	0.40	0.0006	15.2	-	-	-

8.2.2 Alternative site balance scheme

One of the identified problems in the proper optimization of the model has been the large number of loss terms that need to be concurrently optimized while their weighting is also not straightforward. In an attempt to remedy this problem we introduced a different scheme to ensure site balance. Instead of having an additional loss term as in the experiments before, we use a normalization scheme introduced by Nai et al.[34] which implicitly ensures site balance by adding an additional normalizing transformation to the model output. Through this, we also hoped to make the optimization problem better posed. The detailed settings of this run are outlined in table 8.7.

Table 8.7: Settings of the run with a site balance normalization scheme.

Run overview	
Setting	Option
Switch scheme	STE
Normalization scheme	structural enforcement
IC	included
Weights	
Weight	Value
λ_D	1
λ_P	1
λ_{MC}	0.0000888888889
λ_{SR}	0.000001

The corresponding loss curves are shown in figures 8.10 and 8.11 and the corresponding concentration profiles are shown in figure 8.12.

Overall, the loss curves and concentration profiles look very similar to the results obtained from the second reference model. Since the site balance is now enforced through the normalizing transform instead of a separate loss term, there are only four loss terms. Significant differences can only be observed for the concentration profiles where the concentration curve of CO_2 agrees noticeably less with the data. In the adsorbed species, there is somewhat qualitatively better agreement for O^* but the surface coverage for CO^*

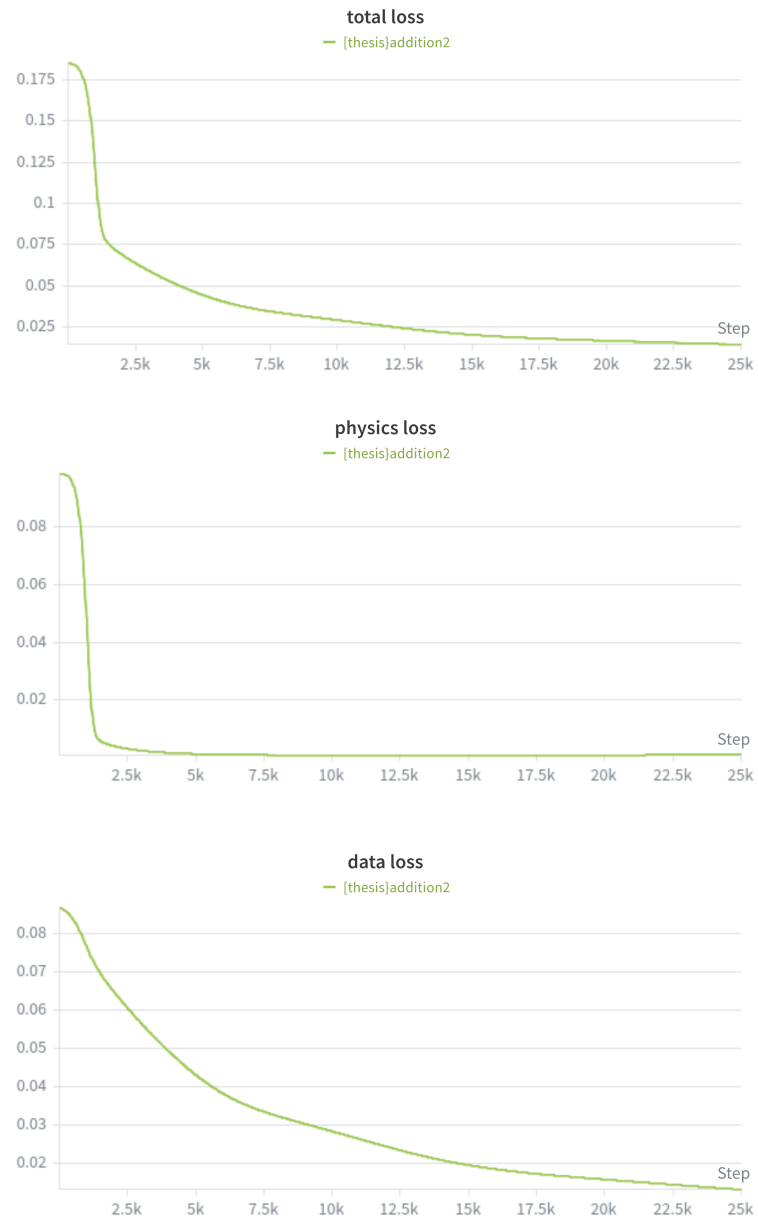


Figure 8.10: Total, physics and data loss curves for training of the the model with a site balance normalization scheme.

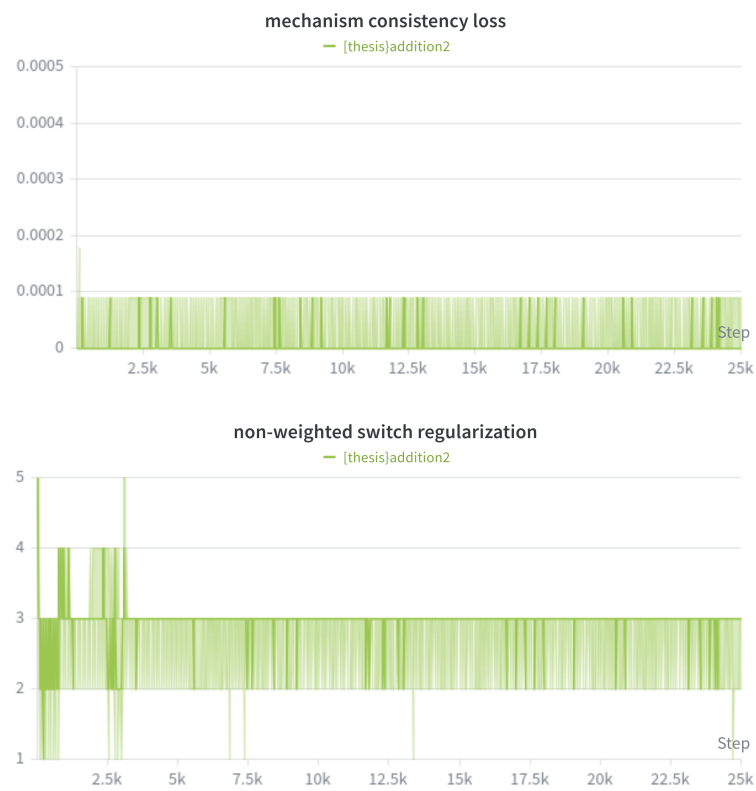
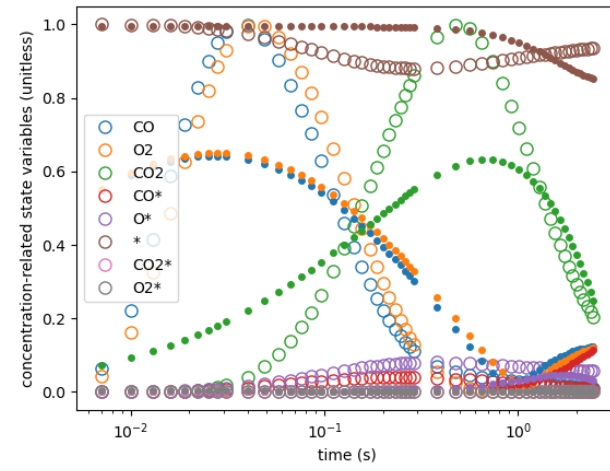
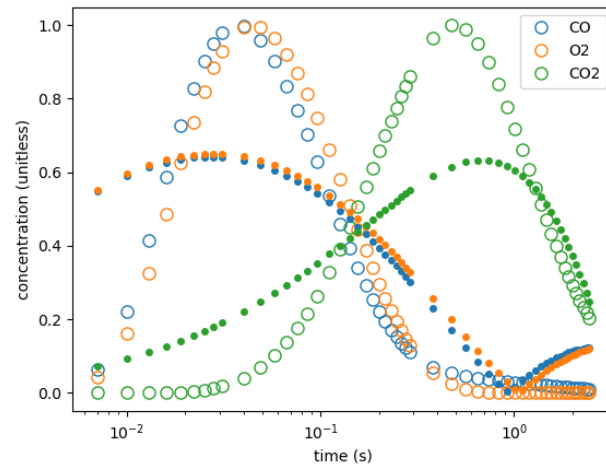


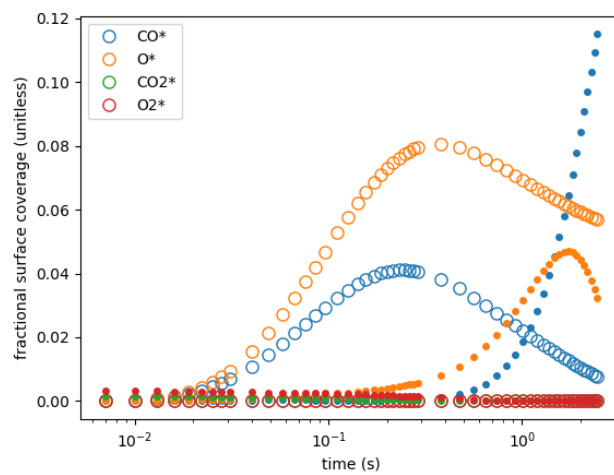
Figure 8.11: SB, MC and SR loss curves for training of the model with a site balance normalization scheme.



(a) All species



(b) Gas species



(c) Surface species (except *)

Figure 8.12: Concentration curves for the model with a site balance normalization scheme.

seems to diverge towards the end of the time series. The predicted switches and rate constants shown in table 8.8 also show very poor agreement with the ground truth. Overall, the normalization scheme for ensuring site balance does not improve the quality of the model despite its theoretical advantages over a separate loss. This suggests that the flaws of the proposed model are deeper such that a single change of a site balance mechanism is not sufficient to obtain proper accuracy.

Table 8.8: True and predicted switches and rate constants of the model with a sigmoid switching function.

Reaction	1	2	3	4	5	6	7	8	9
s_{opt}	0	0	0	0	1	0	1	1	0
s	1	1	1	1	1	1	0	0	0
k_{opt}	0.0014	0.00006	0.039	0.00068	0.065	0.21	0.63	1.11	0.013
k	15.0	0.70	0.01	0.40	0.0006	15.2	-	-	-

Part V

Conclusion

In this work we have investigated the use of PINNs, and more accurately KINNs, to predict chemical reaction mechanism and determine their corresponding rate constants. By concurrent training with the goal of satisfying the underlying physical laws as well as fitting to experimental data we proposed an architecture that, from a range of potential mechanisms picks out the most likely one. The reaction mechanism selection has been implemented in terms of switching functions that allow the model to learn which reactions to turn "on" or "off". Corresponding rate constants could then be determined by evaluating gradients on the predicted concentration values of the model.

We began our analysis by re-implementing the KINN framework from Nai et al. [34] from JAX into Pytorch. While JAX offers good support for low-level optimization, PyTorch provides a more user-friendly and widely adopted workflow that simplifies debugging as well as experimentation. After carefully evaluating and validating the code we could perform experiments to reproduce the results of the original work. The obtained results showed good agreement with the results reported in the original publication confirming that our re-implementation was correct and functional.

Next, we implemented a modified version of the code introduce reaction switches to enable the model to learn to pick the correct mechanism from a provided list of candidate mechanisms. While training seemed to be fairly robust with decent convergence behavior, our models failed to properly reproduce concentration profiles of both gas and adsorbed species. Moreover, the reaction switching functions were not able to reliably predict the correct reaction mechanisms or rate constants. While the inclusion of ICs in accordance with work from Gusmao et al. [17] seemed to make the problem better posed, as indicated by improved convergence behavior, this modification did not lead to a significant improvement of the qualitative results. Additional modifications of the architecture including a different switching function scheme as well as the reduction of concurrent loss terms by employing a final normalizing transformation on the network output did also not improve the model such that decent predictive accuracy could be achieved.

Here, we speculate about the some of the potential reasons for this negative result:

1. There exists a fundamental disconnect between the mechanism optimized by the model and the data: the underlying mechanism of the data is described by a PDE but the optimized mechanism an ODE. The original publication claims that the PDE can accurately be approximated by an ODE and actually requires the formulation as an ODE for its construction. However, this differences causes a disconnect between what the data and physics loss are optimizing for which could be a reason for an overall difficult and ill-posed optimization problem. Especially in the context of adding the mechanism switches this might be one of the reasons our model could not be properly optimized.
2. The loss function of the model contains many sub-losses that constitute concurrent optimization goals. In addition to the four losses discussed in the original publication, we introduced a mechanistic consistency as well as switch regularization loss. Finding appropriate weights to ideally balance the different optimization goals is still a major open challenge in PINNs[38] since the choice of appropriate weights is crucial for the model to properly learn. The rather simple

weight selection scheme employed in this work, in accordance with the original publication[56], seemed to be not sufficient for the training of the proposed model.

3. A self-guiding PINN model as suggested here, that is supposed to determine the correct underlying physics on its own is likely very sensitive and unreliable to train. Since the model has the ability to pick and choose its underlying physics from a set of given mechanism, trying to satisfy all other loss objectives as best as it can, creates a very unstable optimization problem. In this setting it is likely that the model will find a way to minimize all loss objectives by switching off more mechanisms than what is physically sound or reasonable. The results obtained in this work seem to suggest that for PINNs it is crucial to have the physics properly defined beforehand, chosen by an expert in a way that makes physical and chemical sense.

This work highlights that there are remaining major obstacles for the successful application of PINNs and KINNs for the identification of reaction mechanisms and prediction of rate constants. At the time of writing, the overall methodology does not seem appropriate for supporting this goal with particular issues with respect to the choice of loss function weights. The here employed simple procedure of matching loss magnitudes by appropriate weight factors has not lead to the successful training of our model and generally seems insufficient for complex physical systems. More sophisticated approaches have been proposed before[7] including integrated concurrent weight optimization as well as annealing strategies. It remains to be seen if and how much such approaches could benefit the training of the models proposed in this work.

Appendix

Code availability

TAP-KINN

The paper presenting TAP-KINN can be found [here](#), and the code published under an MIT license is available [here](#).

M-KINN

The code can be found [here](#).

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